

Common Phase & Entanglement in Cold-atomic Quantum Field Simulators

Taufiq Murtadho



Theory: Marek Gluza, Nelly Ng

Experiment: Jörg Schmiedmayer, Sebastian Erne, Federica Cataldini, Mohammadamin Tajik



Measurement of total phase fluctuation in cold-atomic quantum simulators

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Quantum Science and Technology



PAPER

Extensive entanglement between coupled Tomonaga–Luttinger liquids in and out of equilibrium

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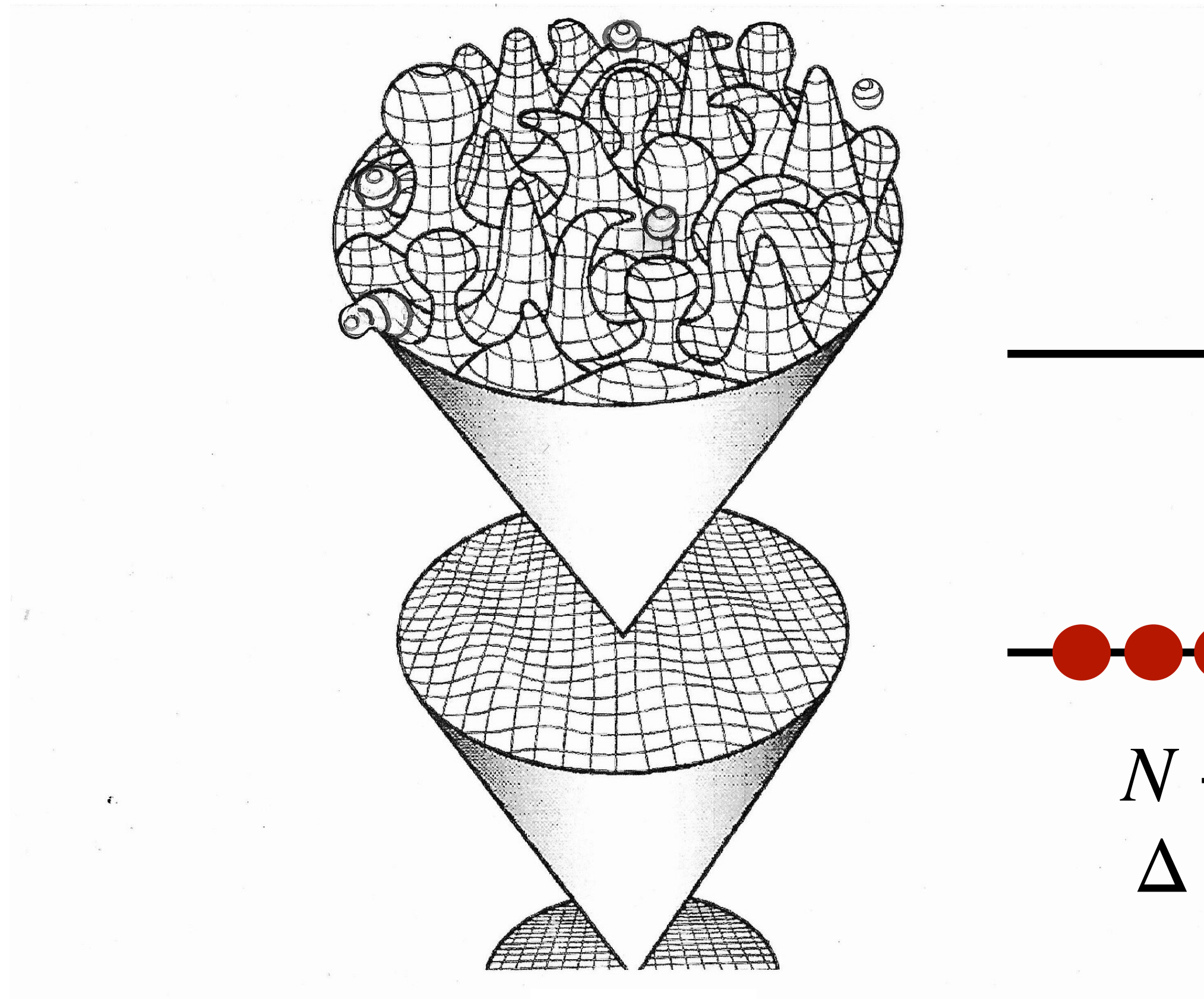
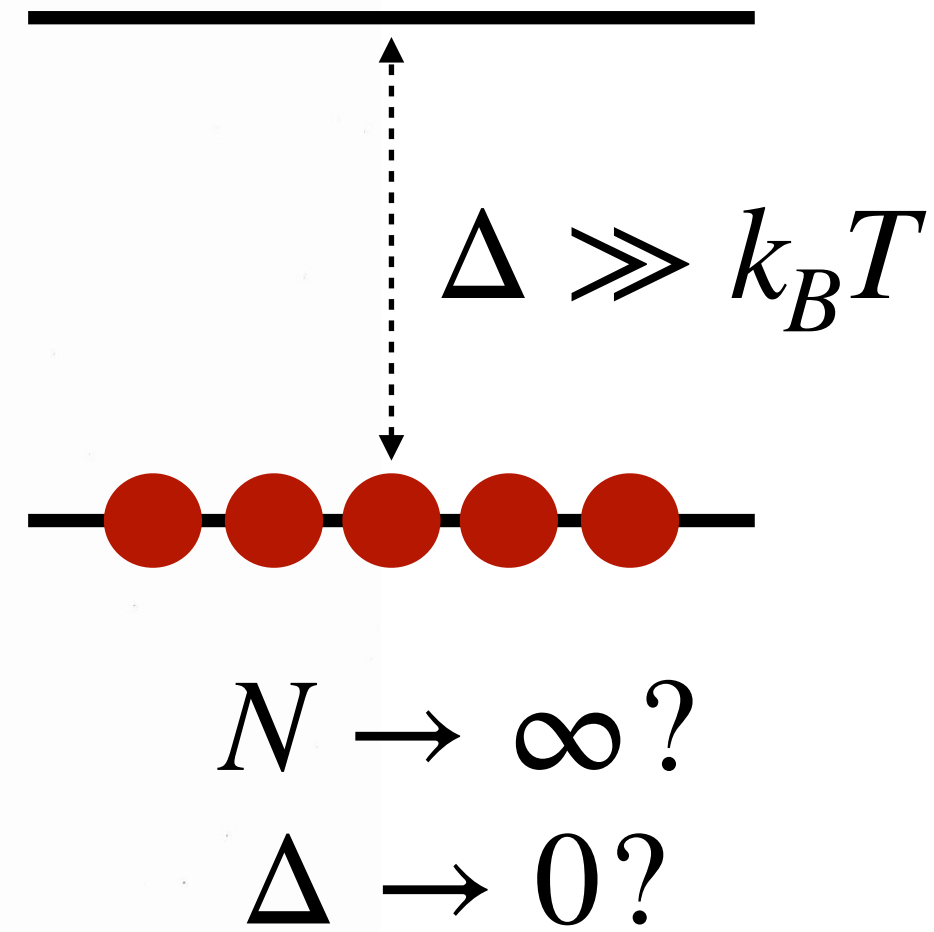


Image: Wikimedia



We do **not** live in a quantum vacuum

Quantum effects live in thermal background

Fluctuation in Bose-Einstein Condensate (BEC)

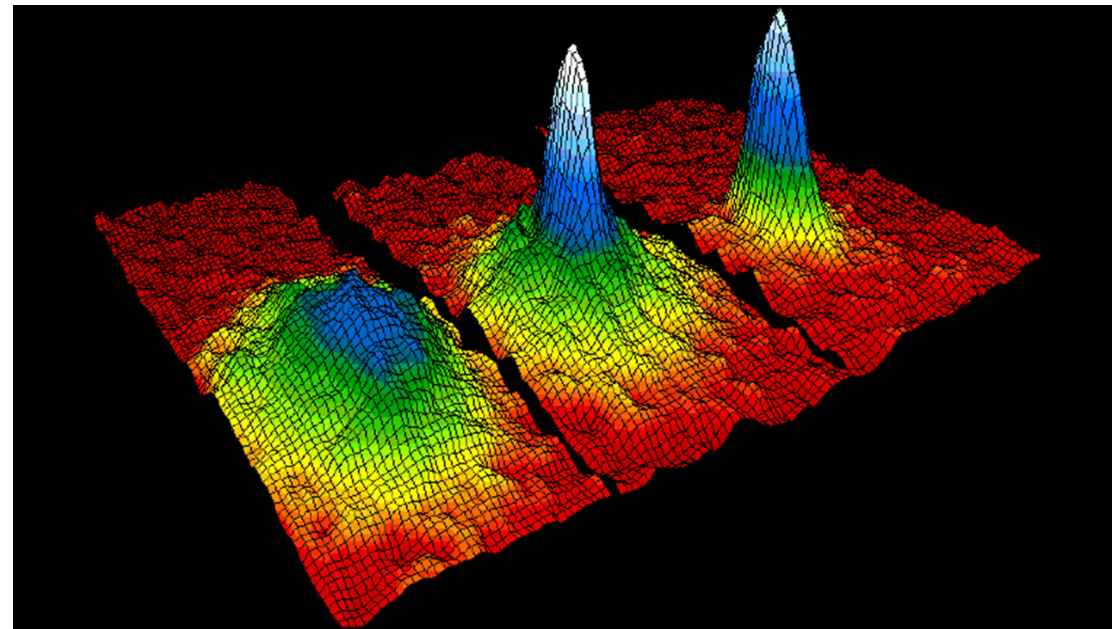


Image: NIST

Anderson, M. H., Ensher, J. R., Matthews, M. R., Wieman, C. E., & Cornell, E. A. *Science*, 269(5221), 198-201 (1995).

Davis, K. B., Mewes, M. O., Andrews, M. R., van Druten, N. J., Durfee, D. S., Kurn, D. M., & Ketterle, W. *PRL*, 75(22), 3969 (1995).

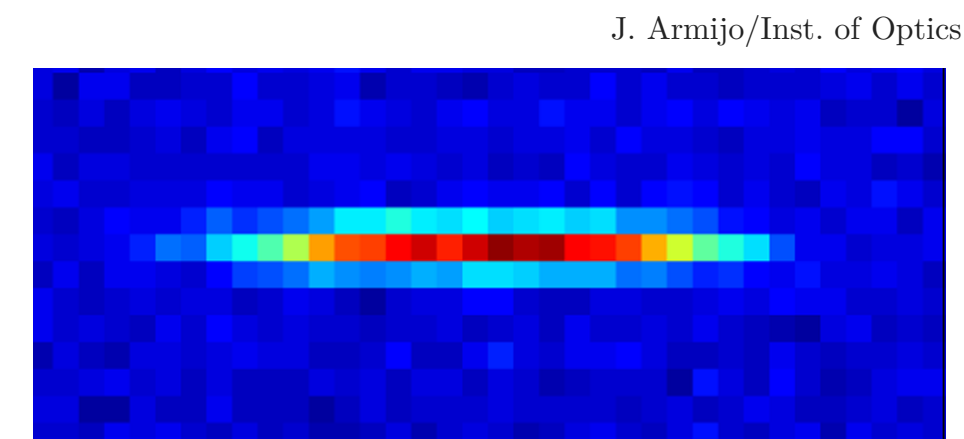


Science Magazine, 22 Dec. 1995

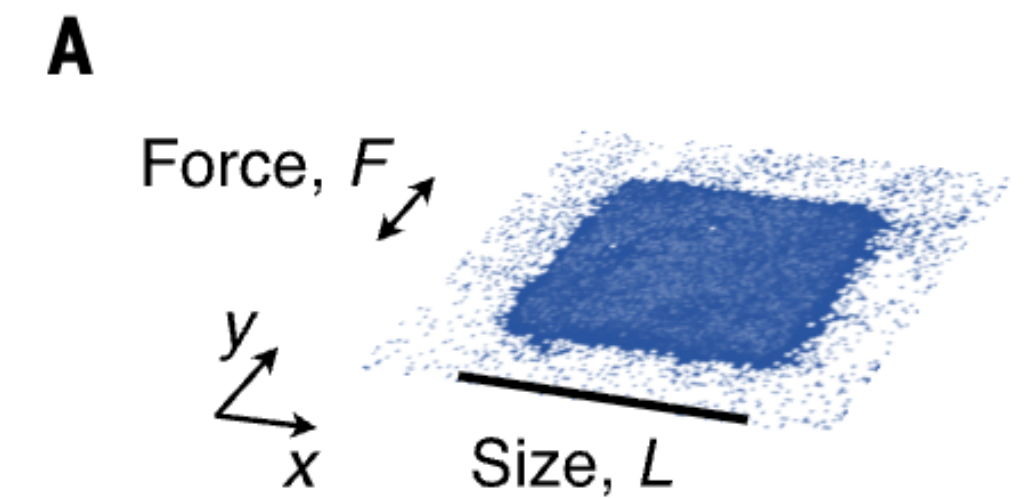
$$\epsilon_k \approx kc$$

$$\epsilon_k \rightarrow 0 \text{ as } k \rightarrow 0 \text{ (Goldstone)}$$

Fluctuation is enhanced in low-dimension (1D/2D)



Armijo, J. *PRL*, 108(22), 225306 (2012).

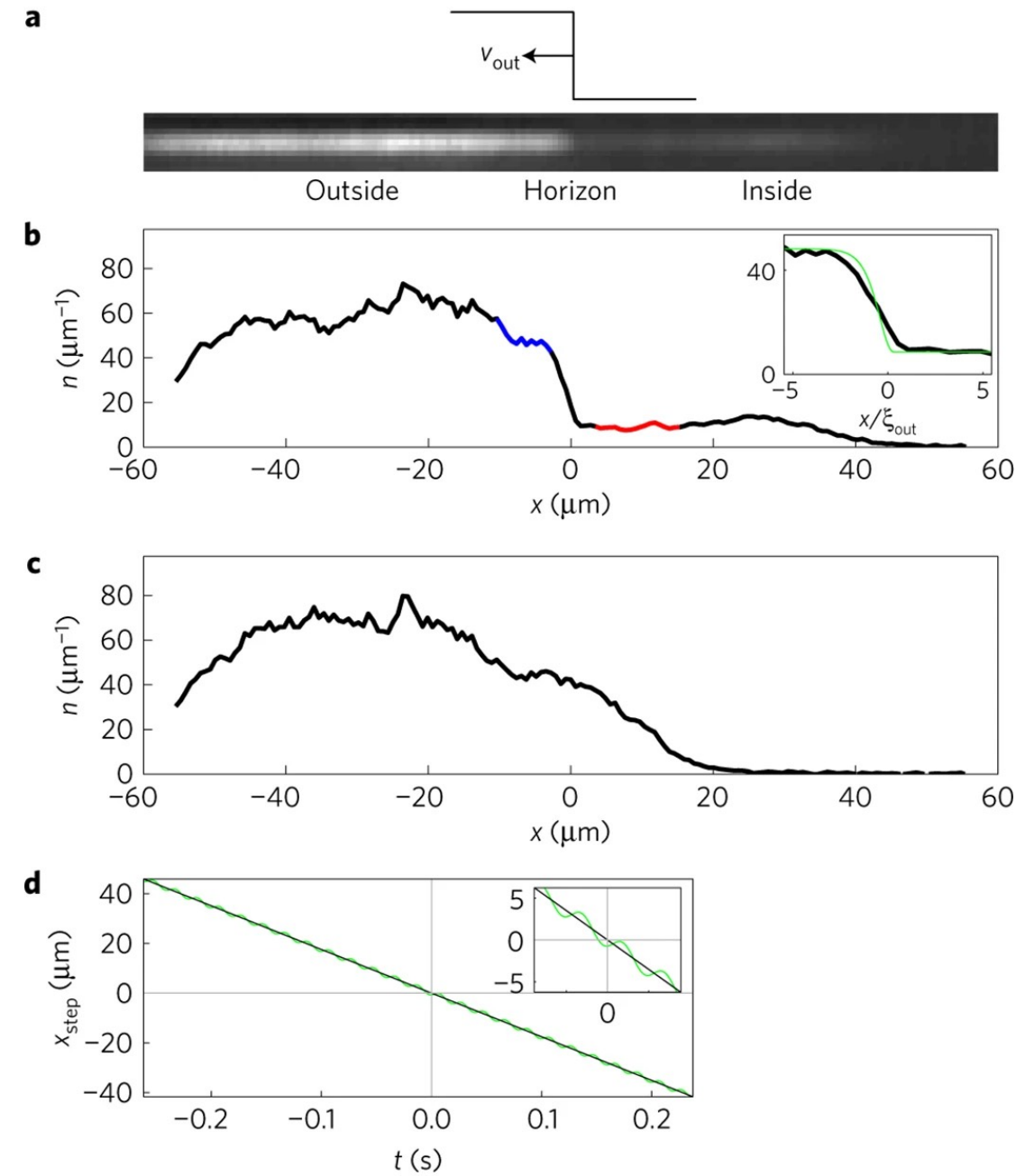


Gazo, M., Karailiev, A., Satoor, T., Eigen, C., Galka, M., & Hadzibabic, Z. *Science*, 389(6762), 802-805 (2025).

$$\int \frac{d^d k}{k^2} \sim \begin{cases} k^{-1} & d = 1 \\ \log k & d = 2 \\ \text{finite} & d = 3 \end{cases}$$

No long-range order at any finite temperature in thermodynamic limit for $d = 1$ and $d = 2$

Quantum Field Simulation with Bosonic Gases

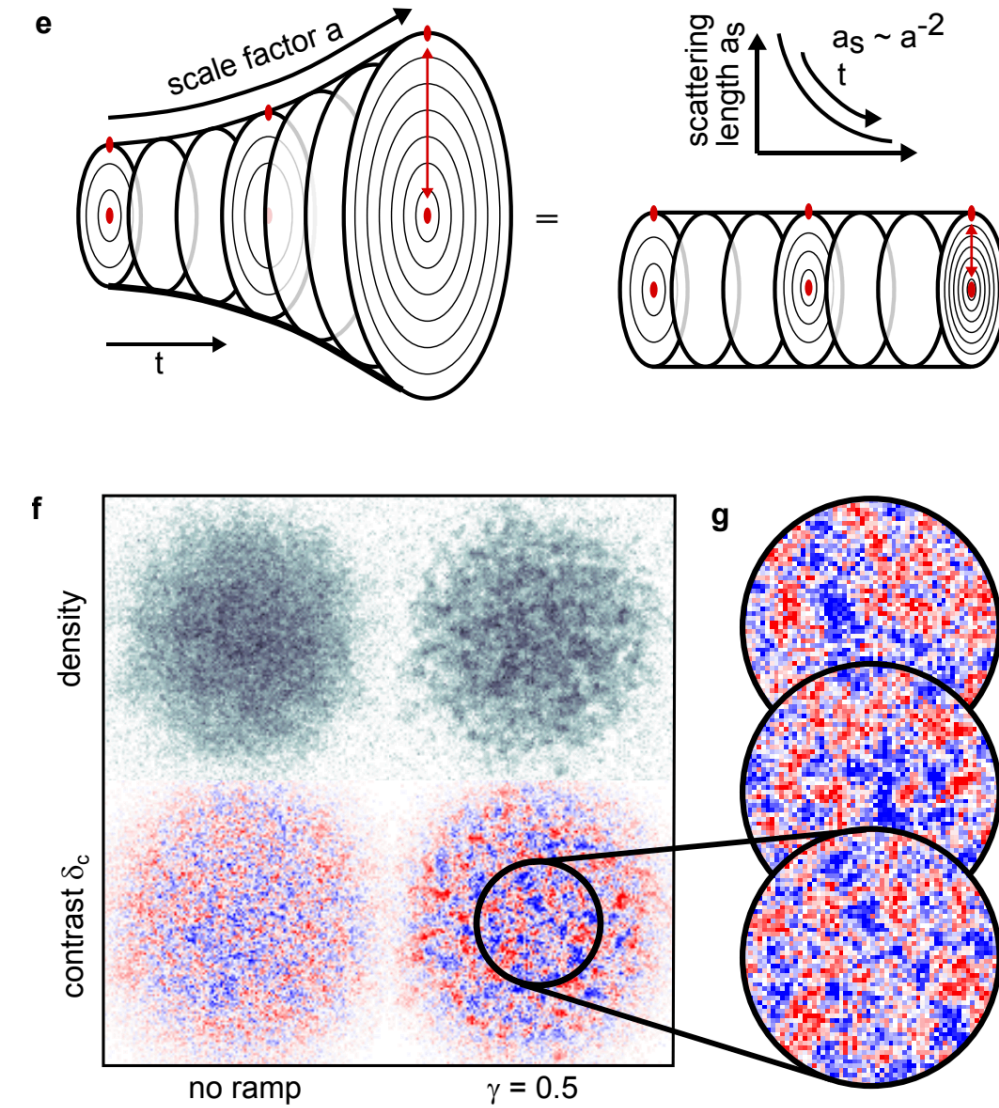


Analogue Black Hole and Hawking Radiation

Garay, L. J., Anglin, J. R., Cirac, J. I., & Zoller, P.
PRA, 63(2), 023611 (2001).

Steinhauer, J. *Nature Physics*, 12(10), 959-965 (2016).

⋮

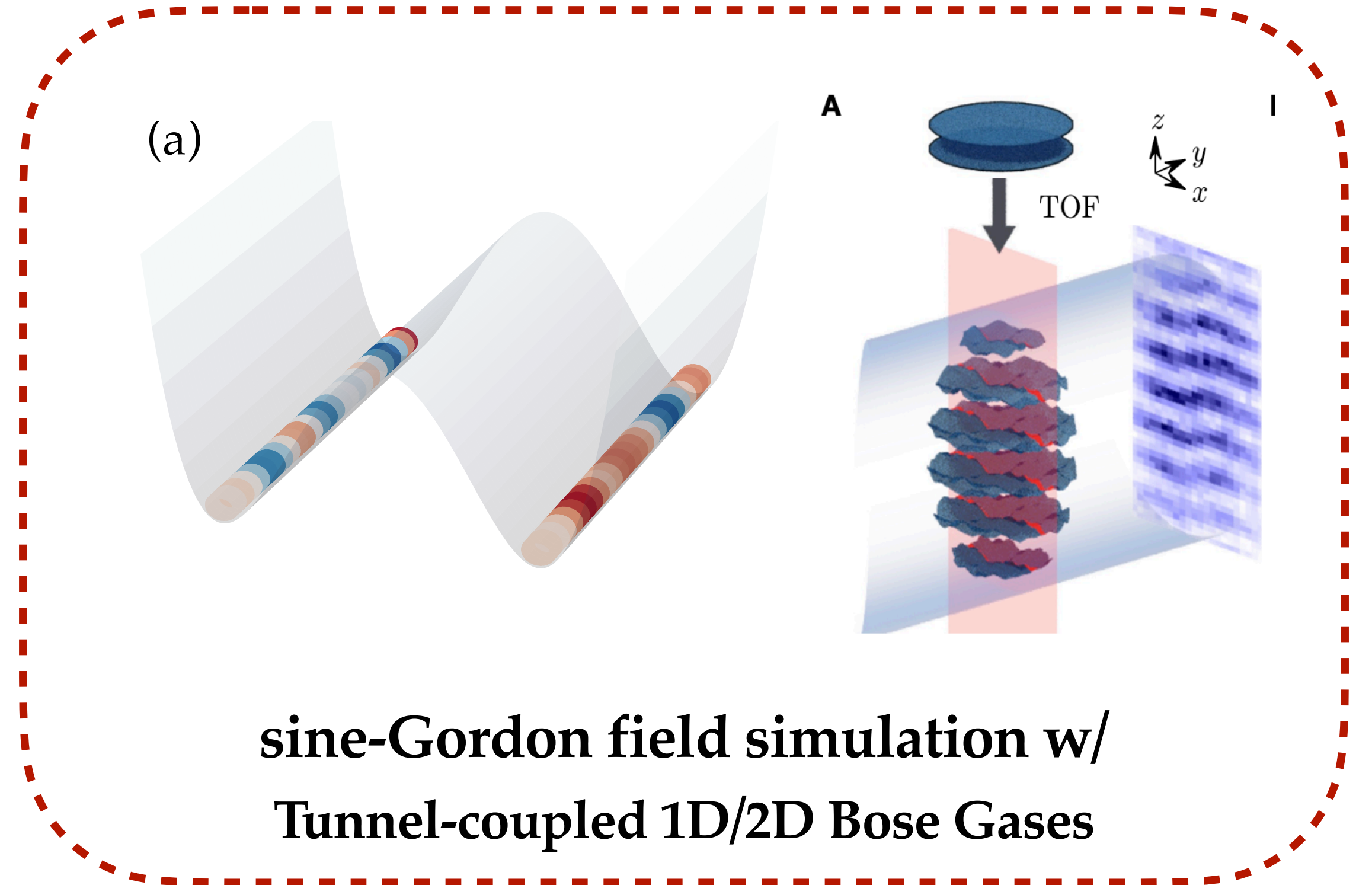


Quantum field in expanding spacetime

Viermann, C., et al. *Nature*, 611(7935), 260-264 (2022).

Agullo, I., Delhom, A., & Parra-López, Á. *PRD*, 110(12), 125023 (2024).

⋮



sine-Gordon field simulation w/ Tunnel-coupled 1D/2D Bose Gases

Tajik, M., et al. *PNAS* 120(21), e2301287120 (2023).

Schweigler, T., et al. *Nature*, 545(7654), 323-326 (2017).

Sunami, S., et al. *PRL*, 128(25), 250402 (2022).

⋮

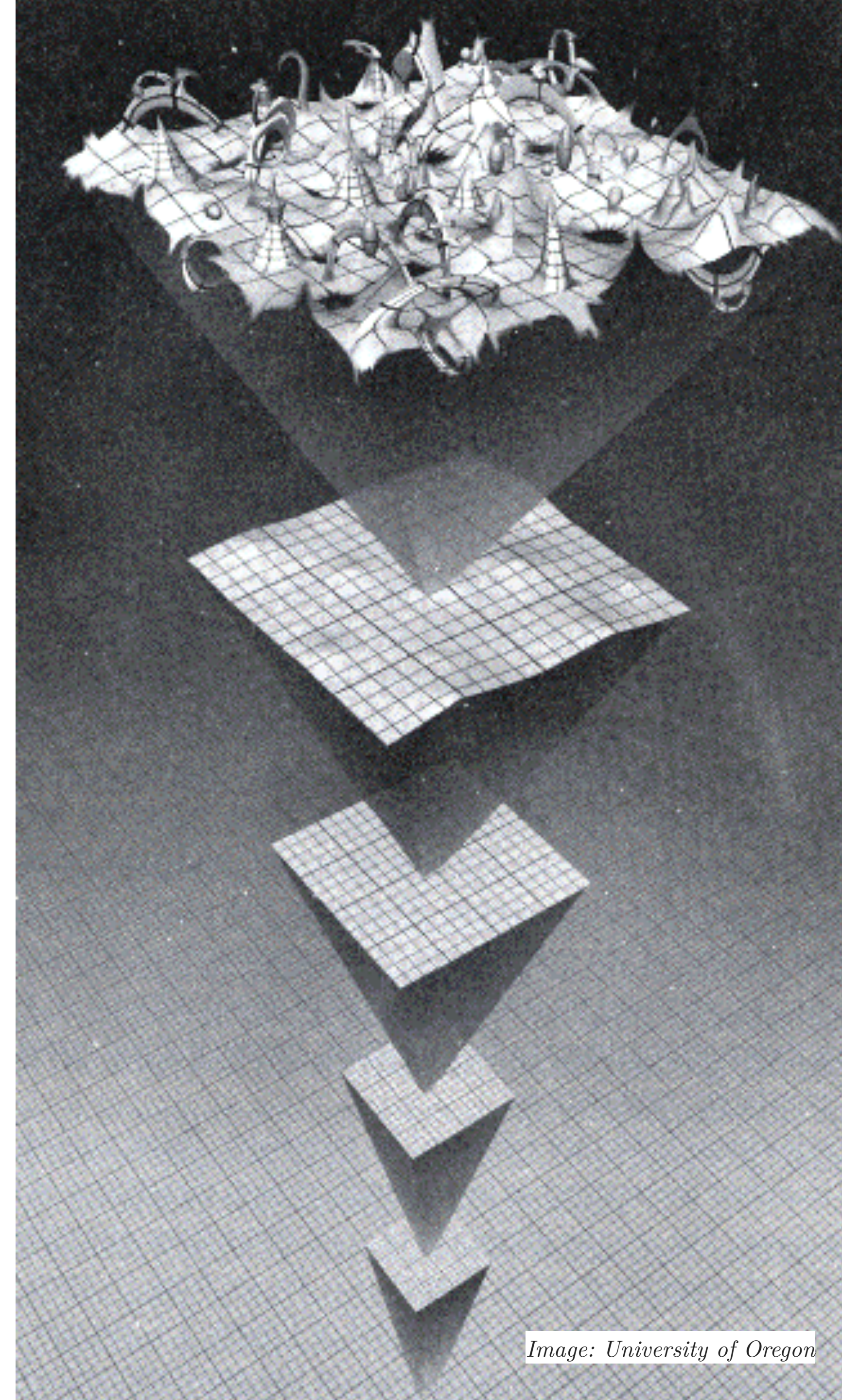
Outline

Motivation

Parallel 1D Bose Gases

Results: Common Phase Measurement
& Entanglement between Parallel 1D Bose Gases

Other Applications & Take-home Messages



Parallel 1D Bose Gases

Image source: Tim Langen's PhD Thesis

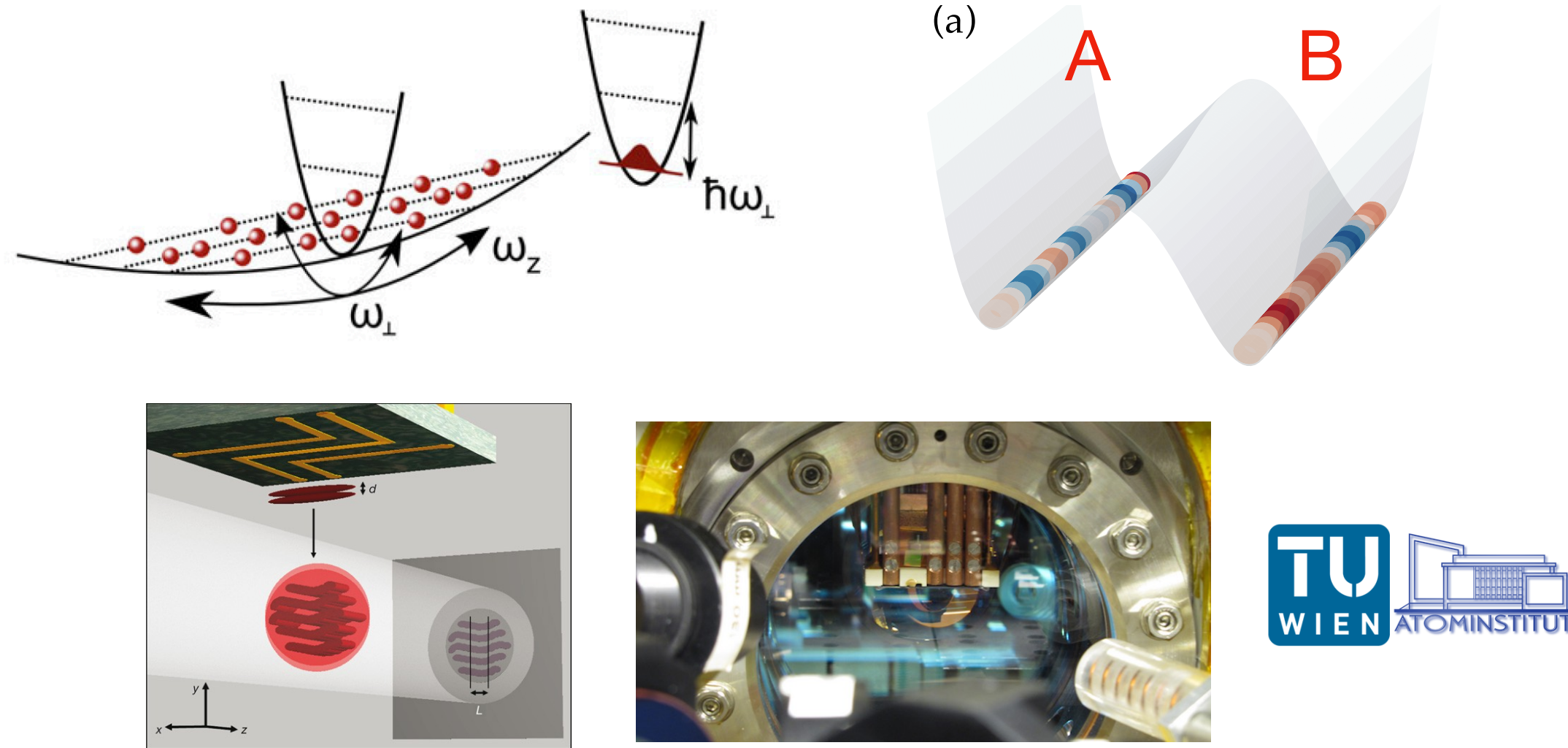


Image source: TU Wien

Collective motional DoF of $\sim 10,000$ atoms,
temperature: **20 - 100 nK**

Bosonic field operator $\hat{\Psi}_{A,B}(z)$ includes *both*
quantum and thermal fluctuations

$$[\hat{\Psi}_{\sigma}(z), \hat{\Psi}_{\kappa}(z')] = \delta(z - z')\delta_{\sigma\kappa}$$

Tunnel-coupled Lieb-Liniger model

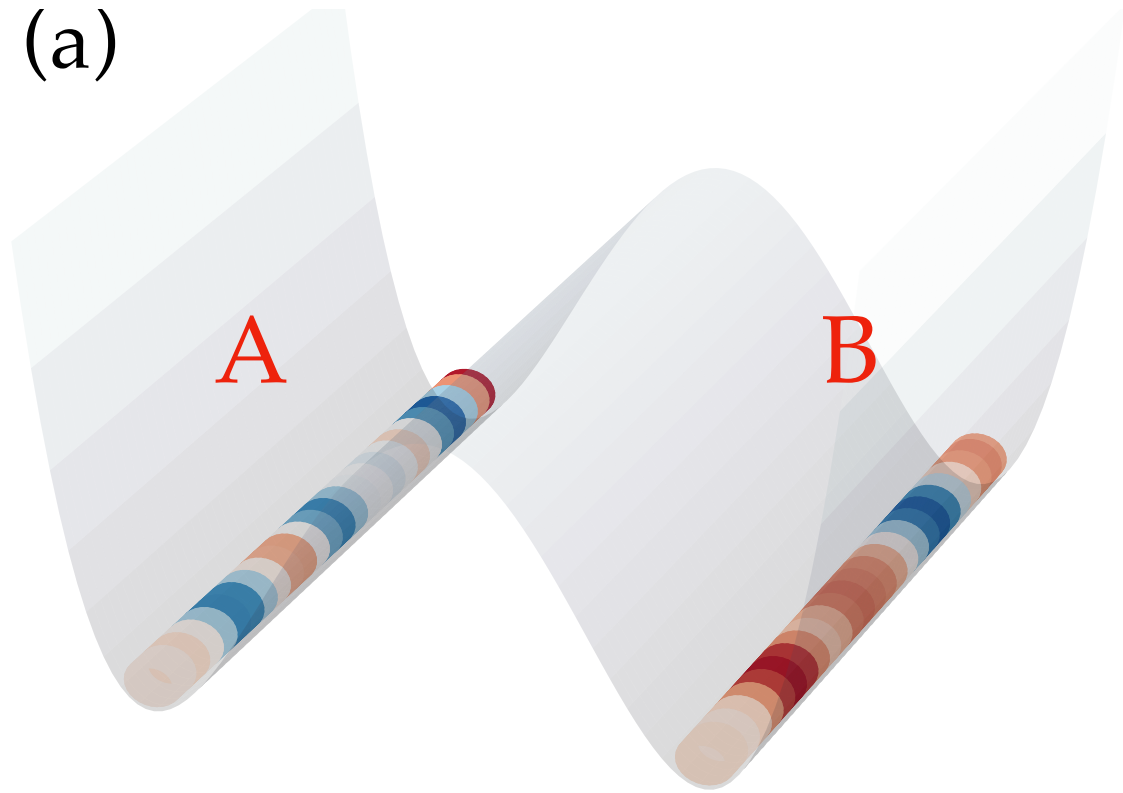
$$\hat{H} = \sum_{\sigma \in \{A,B\}} \left[\underbrace{-\frac{\hbar^2}{2m} \int \hat{\Psi}_{\sigma}^{\dagger} \partial_z^2 \hat{\Psi}_{\sigma}}_{\text{Kinetic + Potential}} + \underbrace{(V(z) - \mu) \int \hat{\Psi}_{\sigma}^{\dagger} \hat{\Psi}_{\sigma}}_{\text{Contact Interaction}} + \underbrace{\frac{g}{2} \int \hat{\Psi}_{\sigma}^{\dagger} \hat{\Psi}_{\sigma}^{\dagger} \hat{\Psi}_{\sigma} \hat{\Psi}_{\sigma}}_{\text{Contact Interaction}} \right] - \hbar J \int dz \left[\hat{\Psi}_A^{\dagger} \hat{\Psi}_B + \text{h.c.} \right]$$

Kinetic + Potential

Contact
Interaction

Tunnel-coupling

Effective Low-Energy Model



$$\hat{\Psi}_\sigma(z) = e^{i\hat{\phi}_\sigma(z)} \sqrt{n_0(z) + \delta\hat{n}_\sigma(z)}$$

$$[\delta\hat{n}_\sigma(z), \hat{\phi}_\kappa(z')] = i\delta(z - z')\delta_{\sigma\kappa}$$

Relative (-) and common (+) fields

$$\delta\hat{n}_\pm(z) = \delta\hat{n}_A(z) \pm \delta\hat{n}_B(z)$$

$$\hat{\phi}_\pm(z) = \hat{\phi}_A(z) \pm \hat{\phi}_B(z)$$

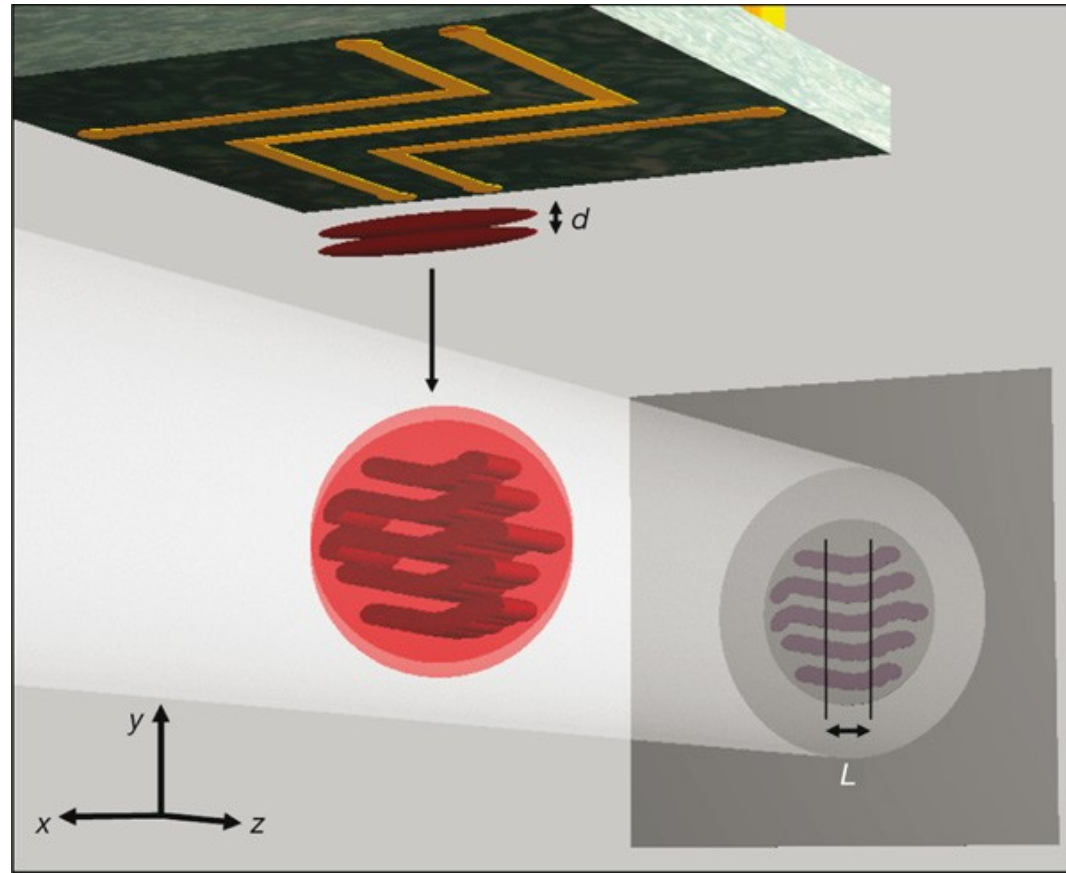
$$\hat{H} \approx \hat{H}_{SG} = \hat{H}_{TLL}[\delta\hat{n}_+, \hat{\phi}_+] + \hat{H}_{TLL}[\delta\hat{n}_-, \hat{\phi}_-] - 2\hbar J \int n_0(z) \cos \hat{\phi}_-(z) dz$$

$$\hat{H}_{TLL}[\delta\hat{n}, \hat{\phi}] = \frac{\hbar c}{2} \int \left[\frac{\pi}{K} (\delta\hat{n}(z))^2 + \frac{K}{\pi} (\partial_z \hat{\phi}(z))^2 \right] dz$$

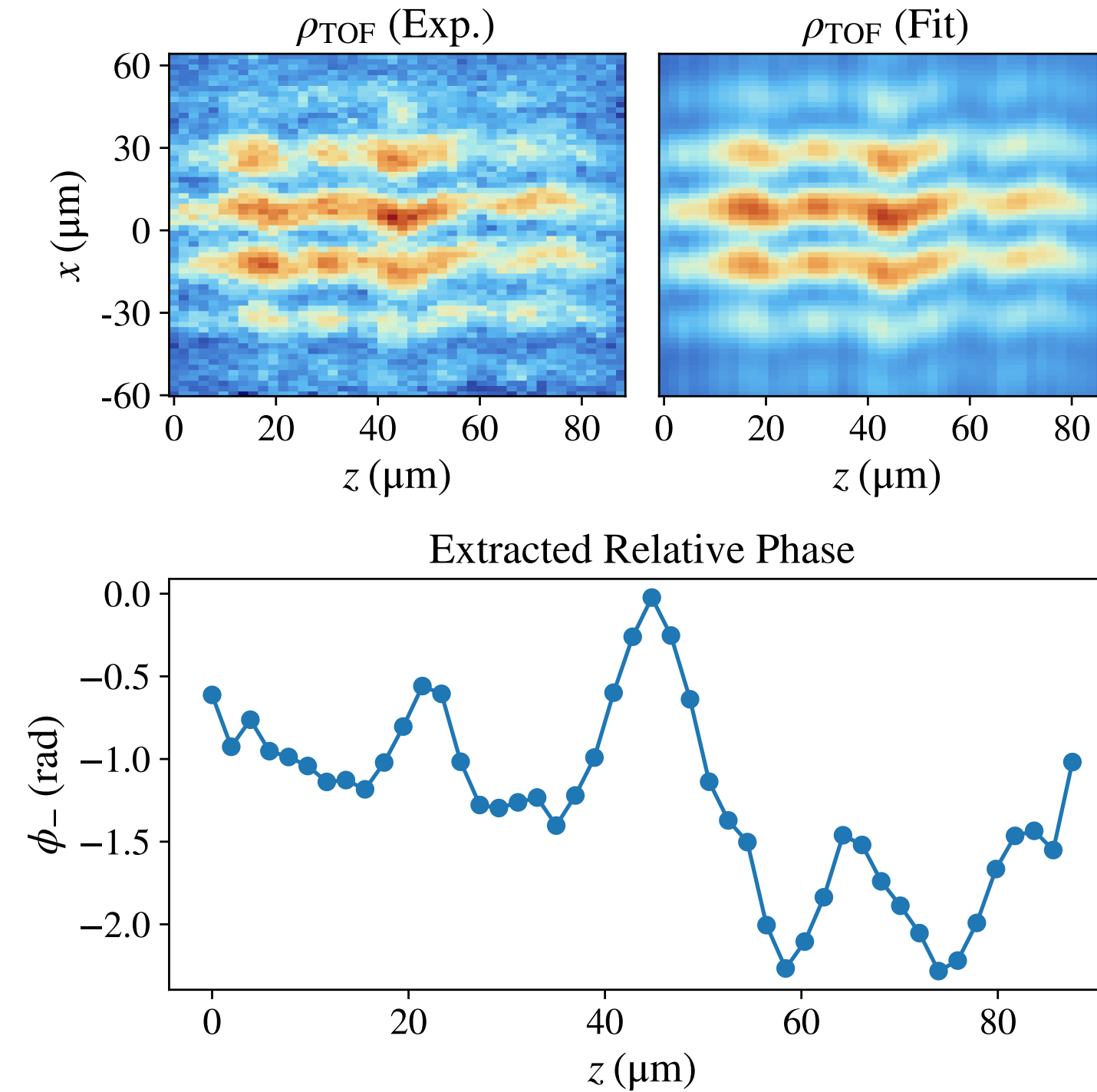
**sine-Gordon
interaction**

Tomonaga-Luttinger Liquid (TLL)

Matter-wave interference readout



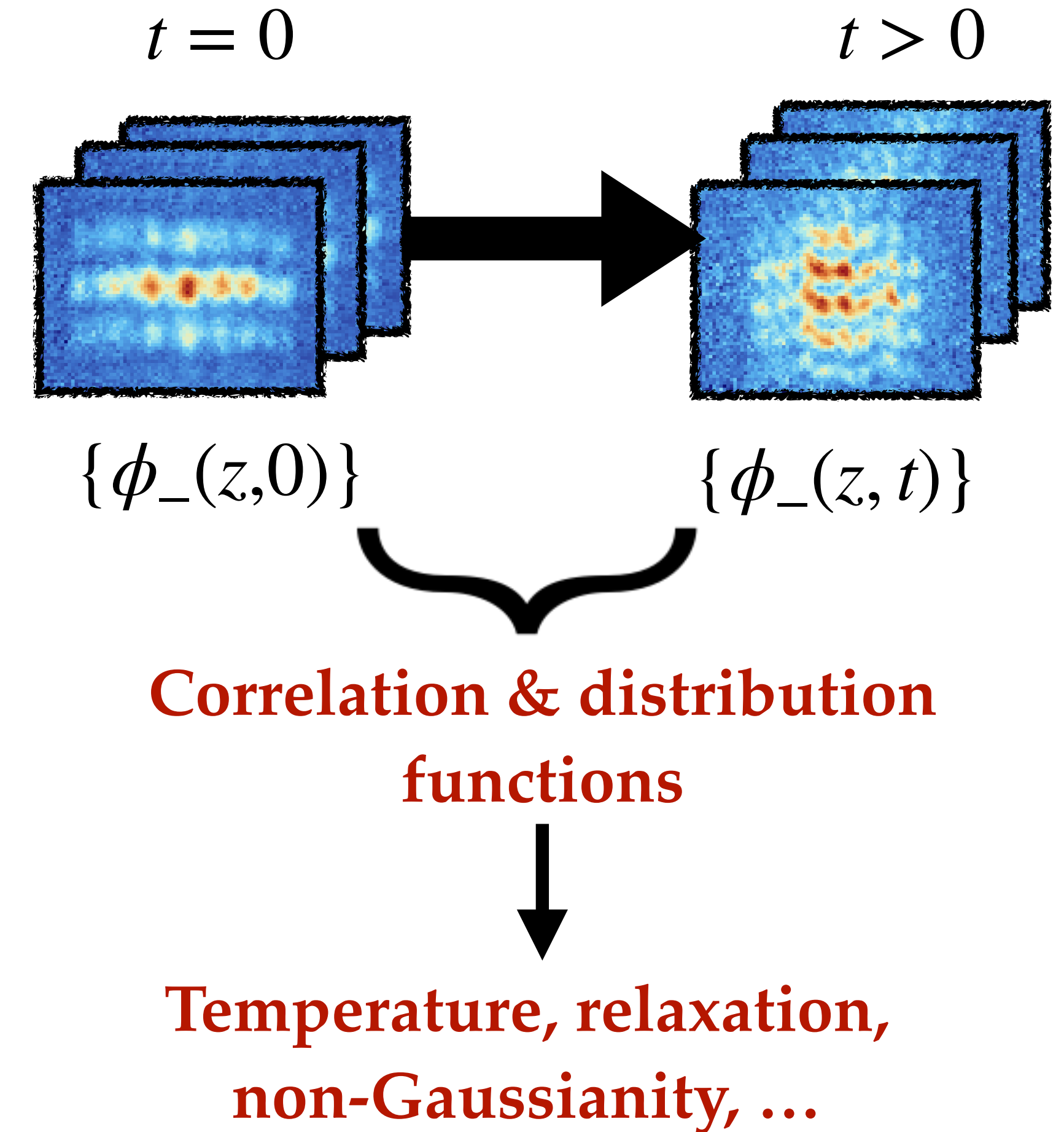
Hofferberth, S., Lesanovsky, I., Fischer, B., Schumm, T., & Schmiedmayer, J.
Nature, 449(7160), 324-327. (2007)



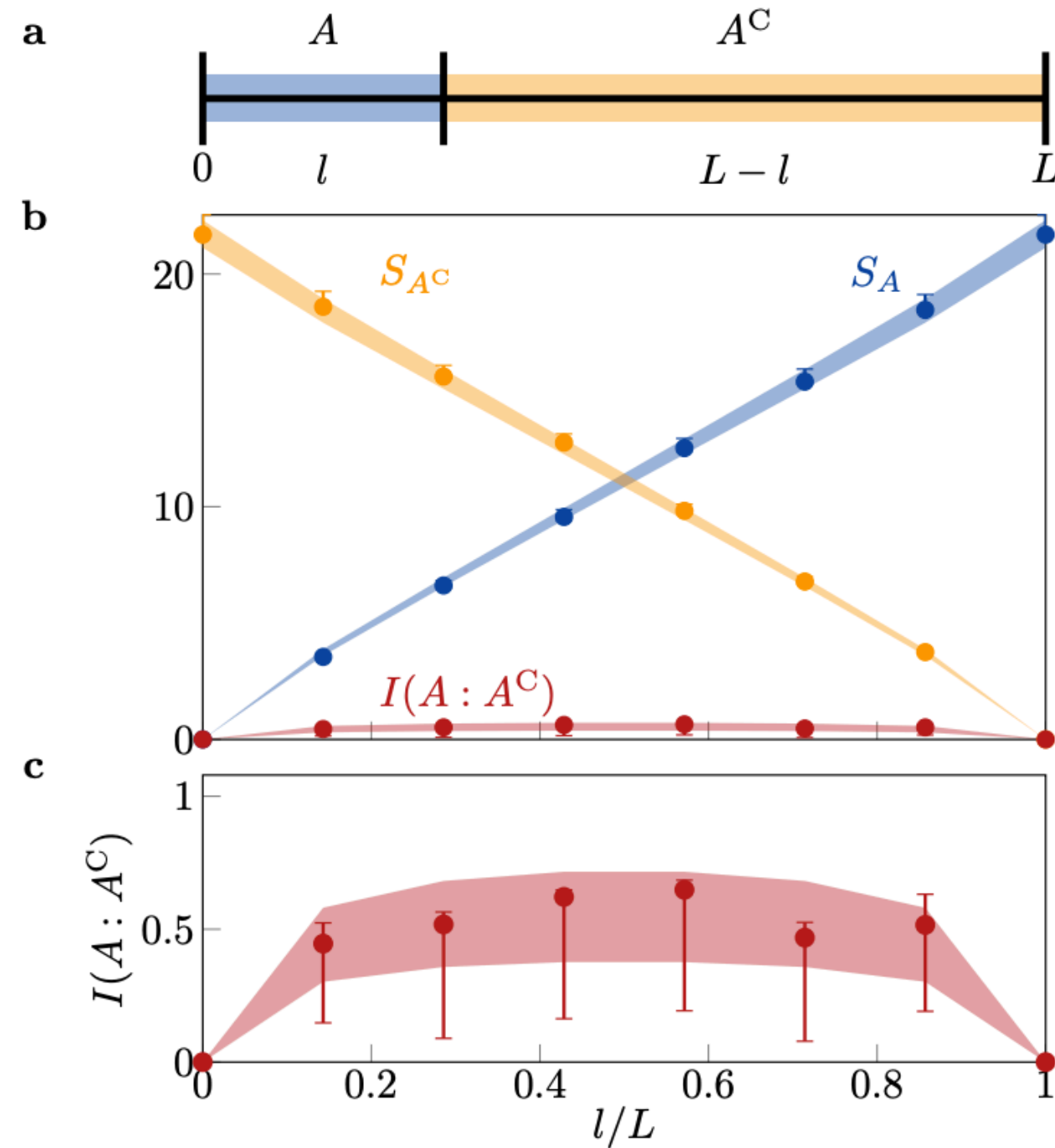
$$\rho_{\text{TOF}}(x, z, \tau) = A(z)e^{-x^2/\sigma_\tau^2} \left[1 + B(z) \cos(kx + \phi_-(z)) \right]$$

van Nieuwkerk, Y. D, Schmiedmayer, J. and Essler, F. *SciPost Physics* 5.5: 046 (2018).

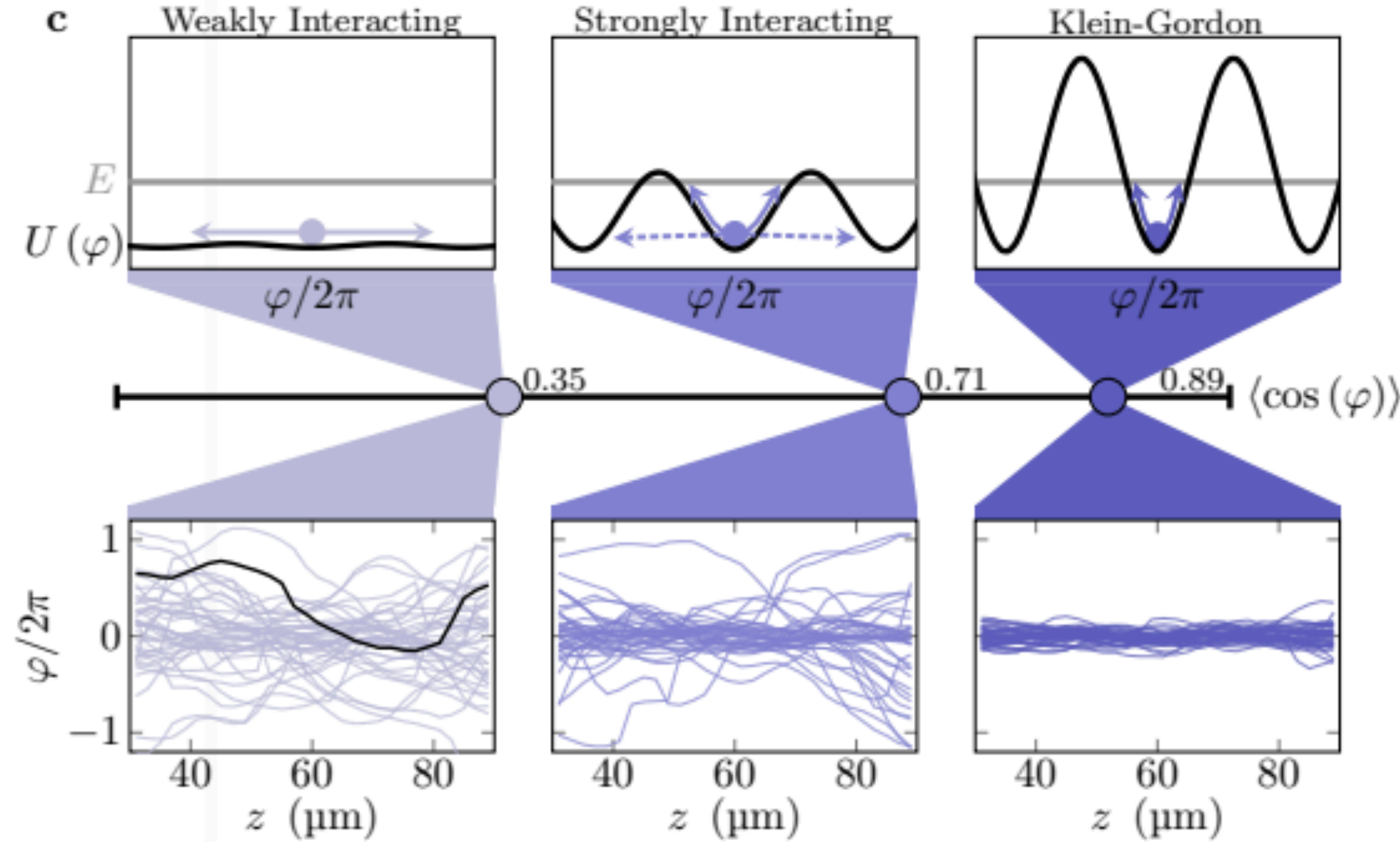
Murtadho, T., et al. *SciPost Physics* 18.2: 065 (2025).



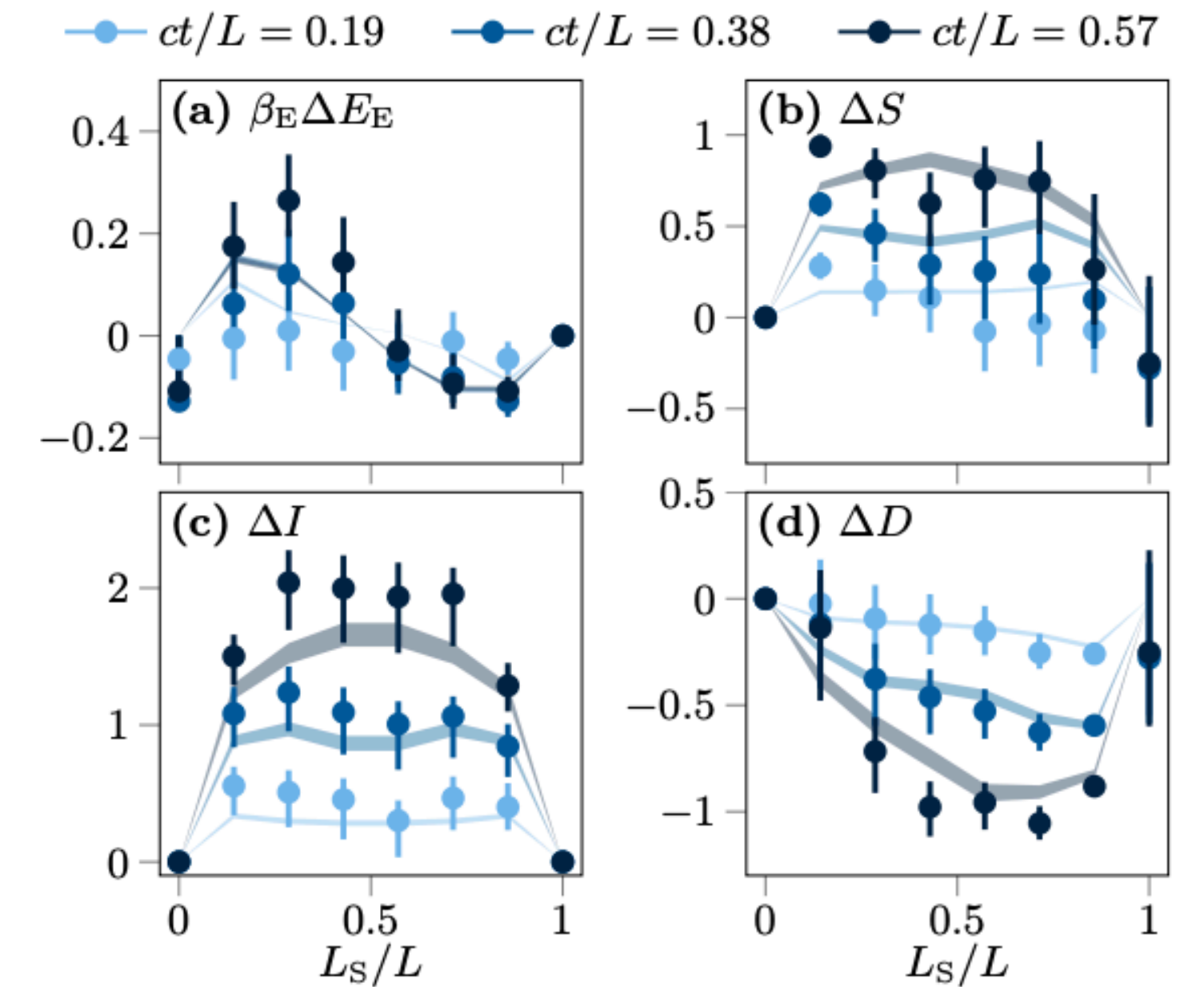
Field theory ft. information theory



Tajik, M., et al. *Nat. Phys.*, 19(7), 1022-1026 (2023).



Jarema, M. T., Tajik, M., Schmiedmayer, J., Weinfurtner, S., & Haas, T. *arXiv:2510.13783* (2025).



Aimet, S., et al. *Nat. Phys.*, 21(8), 1326-1331 (2025).



Keep alive
quantum mechanics!

Entanglement??

Mathé, J., et al. *arXiv:2604.01157* (2026).

Dupays, L., et al. *arXiv:2608.05968* (2026).

Murtadho, T., Gluza, M., & Ng, N. H. Y. *Quantum Sci. Technol.* 11: 035060 (2026)

Measurement bottleneck

- * Limited available measurements create a bottleneck to the phenomena that can be observed
- * Research aim: expand the measurement toolbox and use it to probe quantum phenomena (e.g. entanglement) in quantum field simulators



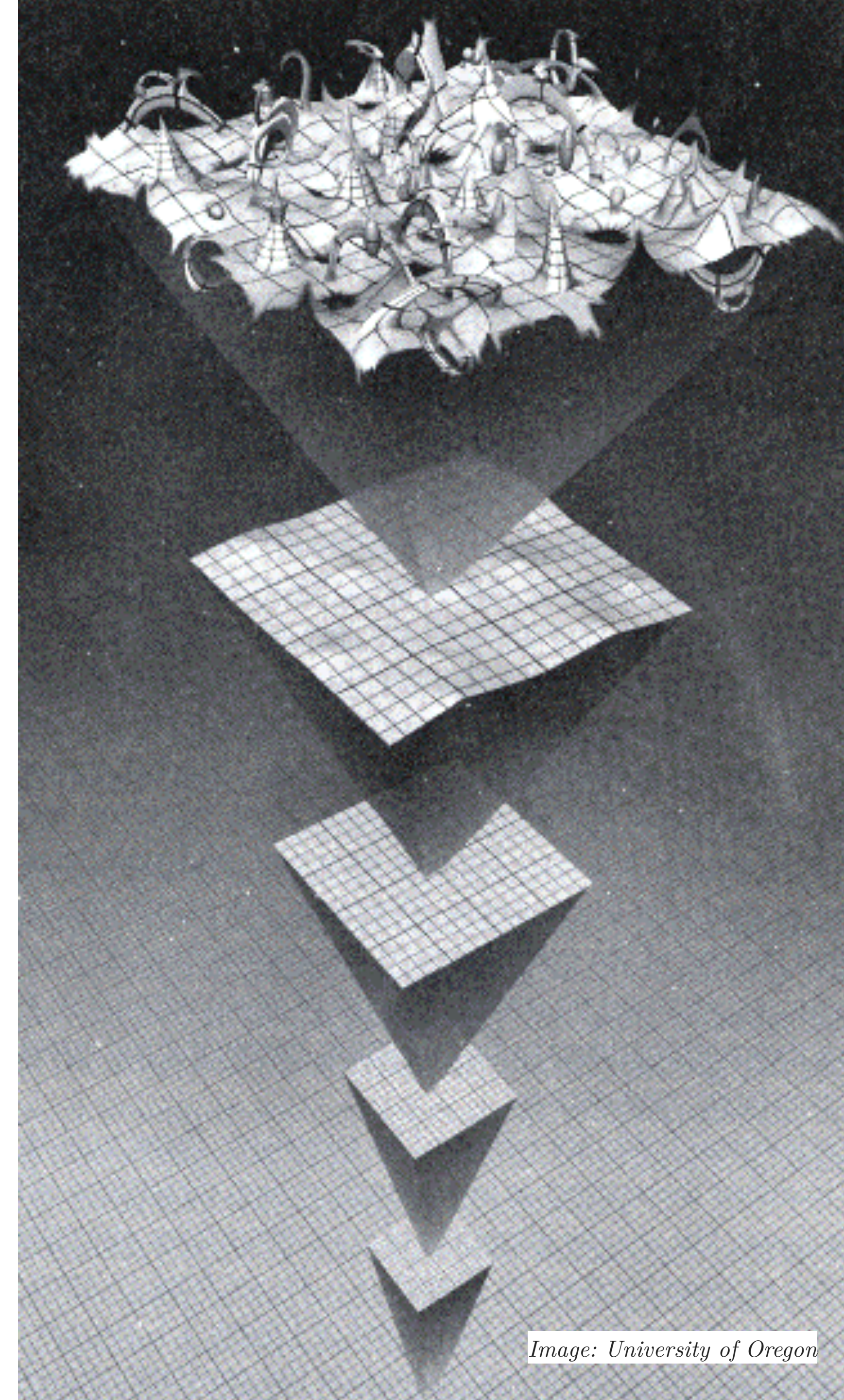
Outline

Motivation

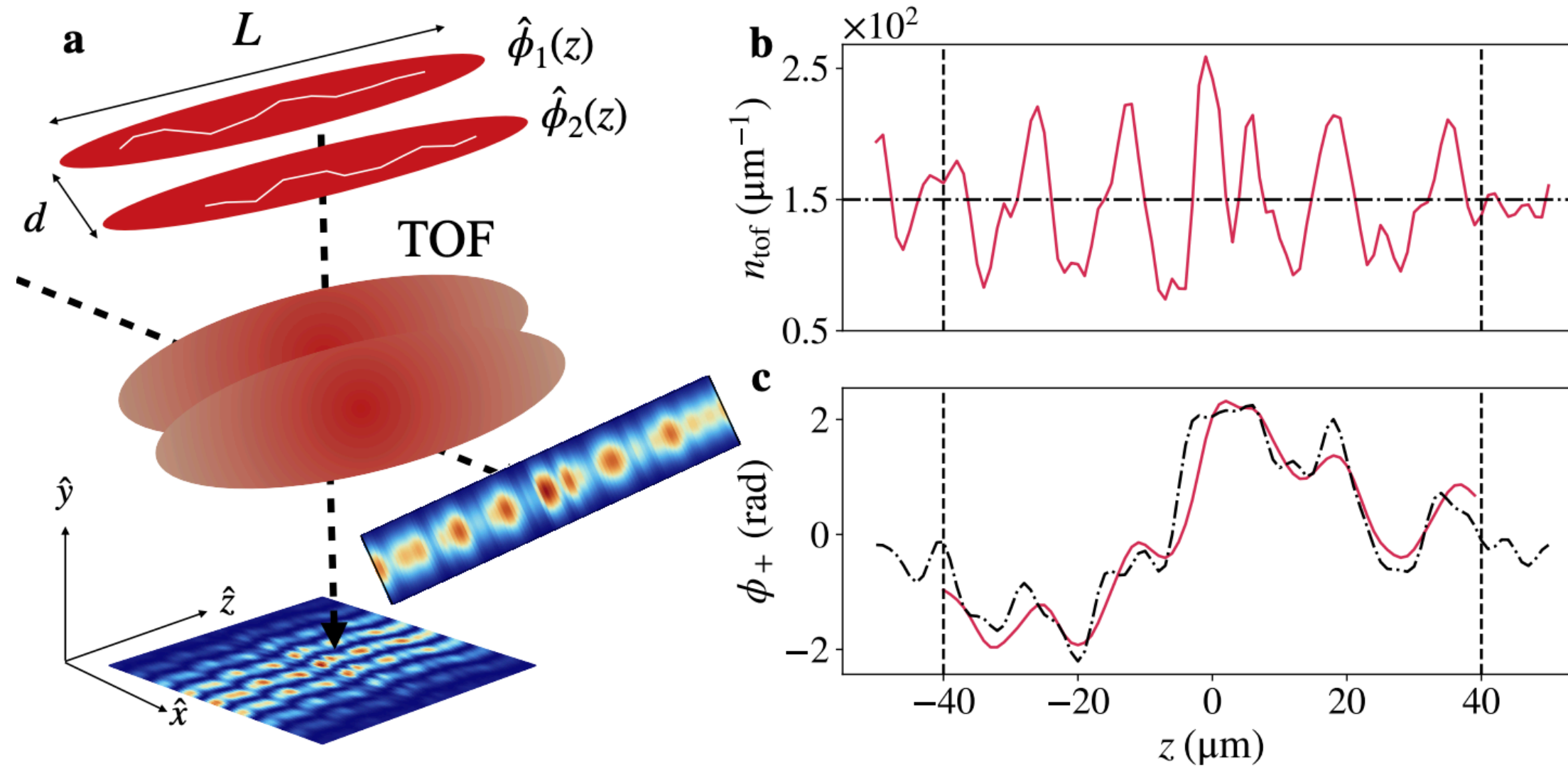
Parallel 1D Bose Gases

Results: Common Phase Measurement
& Entanglement between Parallel 1D Bose Gases

Other Applications & Take-home Messages



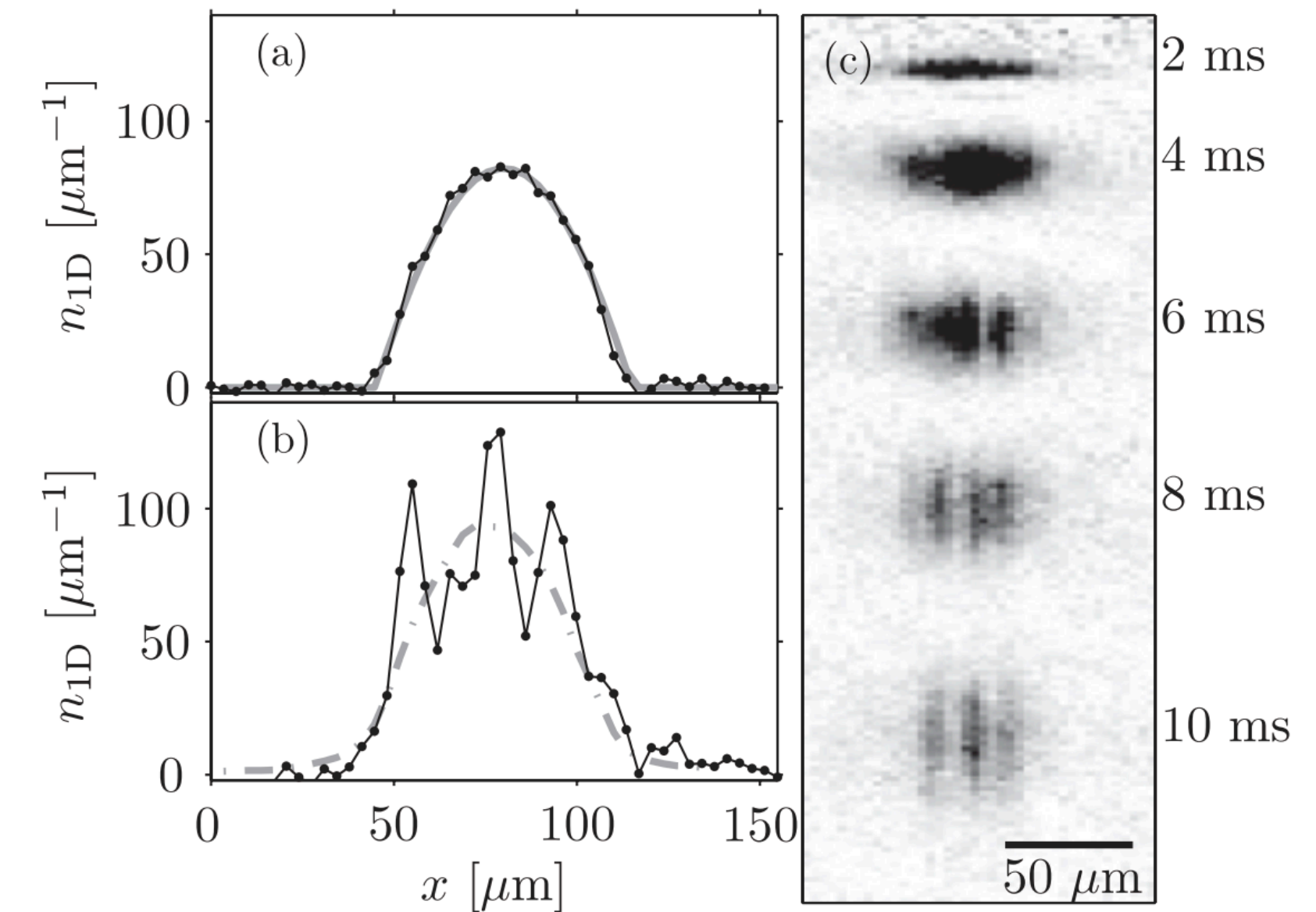
Extracting common phase



Murtadho, T., Cataldini, F., Erne, S., Gluza, M., Tajik, M., Schmiedmayer, J., & Ng, N. *PRR*, 7(2), L022031 (2025).

$$\hat{\phi}_+(z) = \hat{\phi}_1(z) + \hat{\phi}_2(z)$$

$$\partial_z^2 \hat{\phi}_+ \approx \frac{1}{\ell_\tau^2} \left(1 - \frac{n_{\text{tof}}(z, \tau)}{n_0(z)} \right)$$



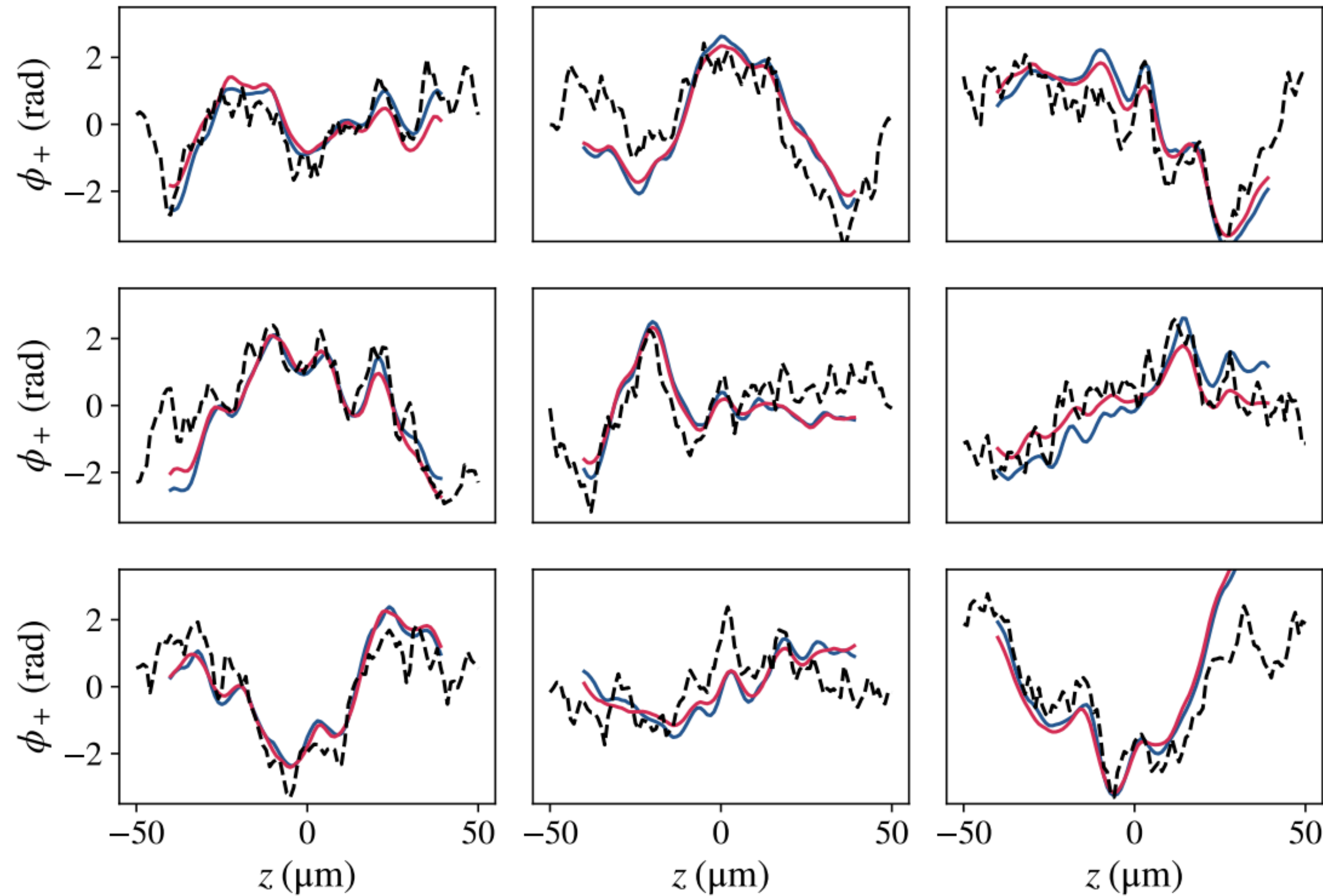
Dettmer, S., et al. *PRL* 87(16), 160406 (2001).

Manz, S., et al. *PRA*, 81(3), 031610 (2010).

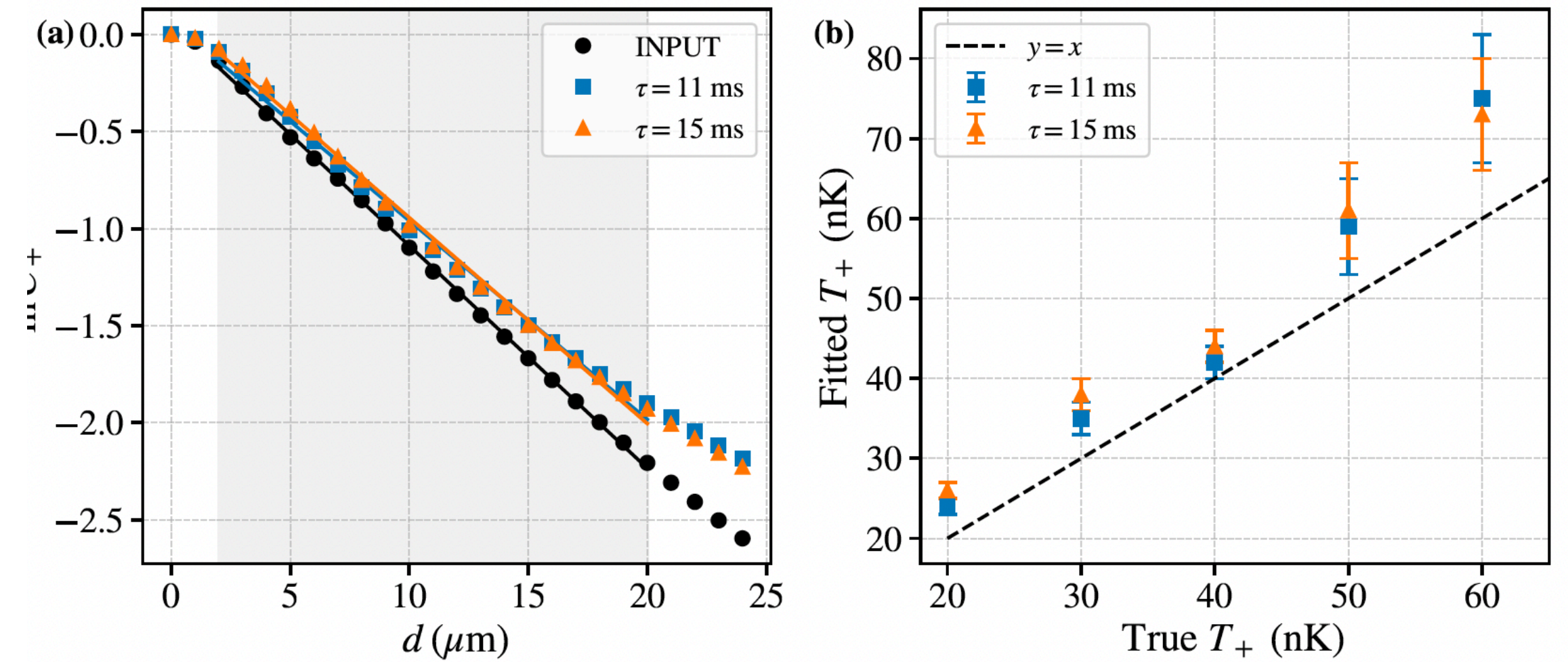
$$\hat{\phi}(z) \rightarrow \delta \hat{n}(z, \tau)$$

$$\partial_\tau \hat{n} + \frac{\hbar}{2m} \nabla \cdot (n_0 \nabla \hat{\phi}) = 0$$

Reconstructing Common Phase Correlation



Vertex correlation thermometry



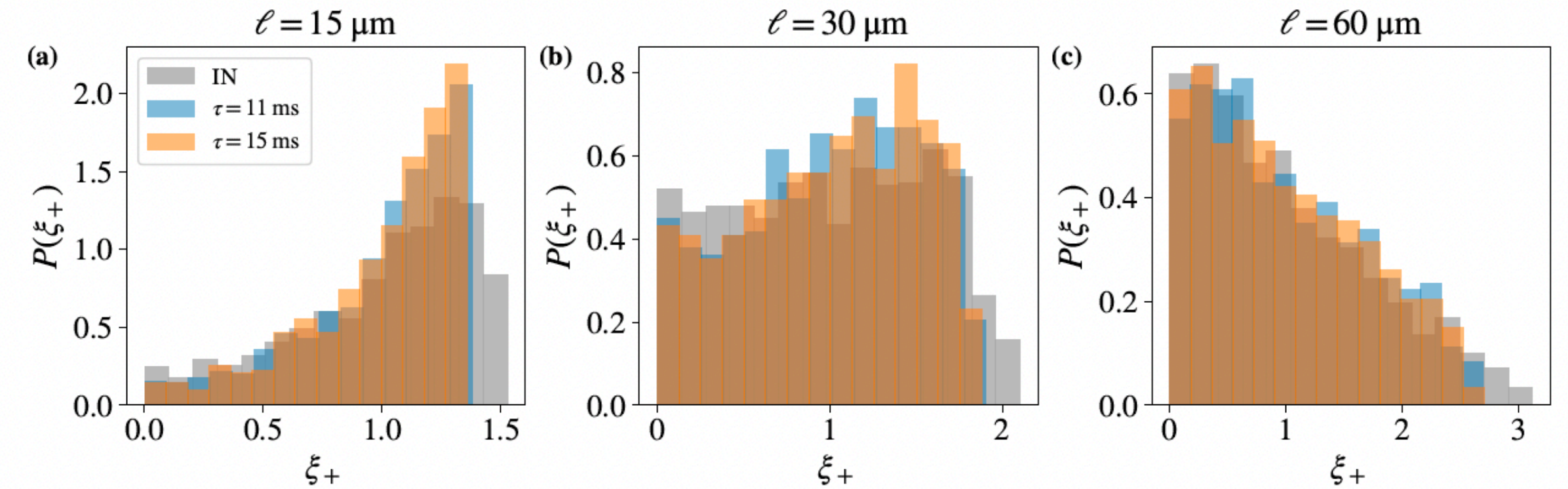
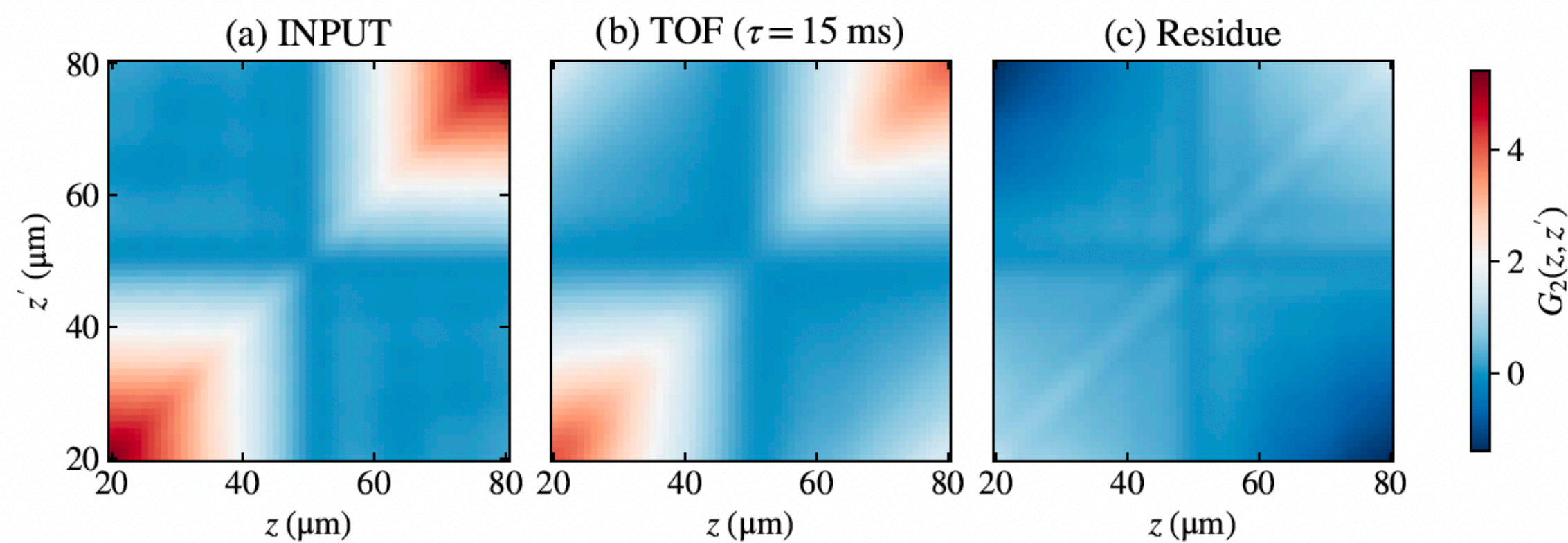
$$C_+(z, z') = \left\langle \exp [i(\phi_+(z) - \phi_+(z'))] \right\rangle$$

$$\sim \exp(-|z - z'|/\lambda_{T_+})$$

$$\lambda_{T_+} = \frac{\hbar^2 n_0}{mk_B T_+}$$

Murtadho, T., Cataldini, F., Erne, S., Gluza, M., Tajik, M., Schmiedmayer, J., & Ng, N. *PRR*, 7(2), L022031 (2025).

Reconstructing Common Phase Correlation



Contrast Distribution Function

$$\xi_+(\ell) = \frac{\left| \int_{-\ell/2}^{\ell/2} \exp(i\phi_+(z)) dz \right|^2}{\left\langle \left| \int_{-\ell/2}^{\ell/2} \exp(i\phi_+(z)) dz \right|^2 \right\rangle}$$

Two-point function

$$G_+^{(2)}(z, z') = \langle [\phi_+(z) - \phi_+(0)][\phi_+(z') - \phi_+(0)] \rangle$$

Can be generalized to higher-order, i.e. $G_+^{(N)}(\mathbf{z})$

Gluza, M., et al. *Commun. Phys.* 3.1: 12 (2020).

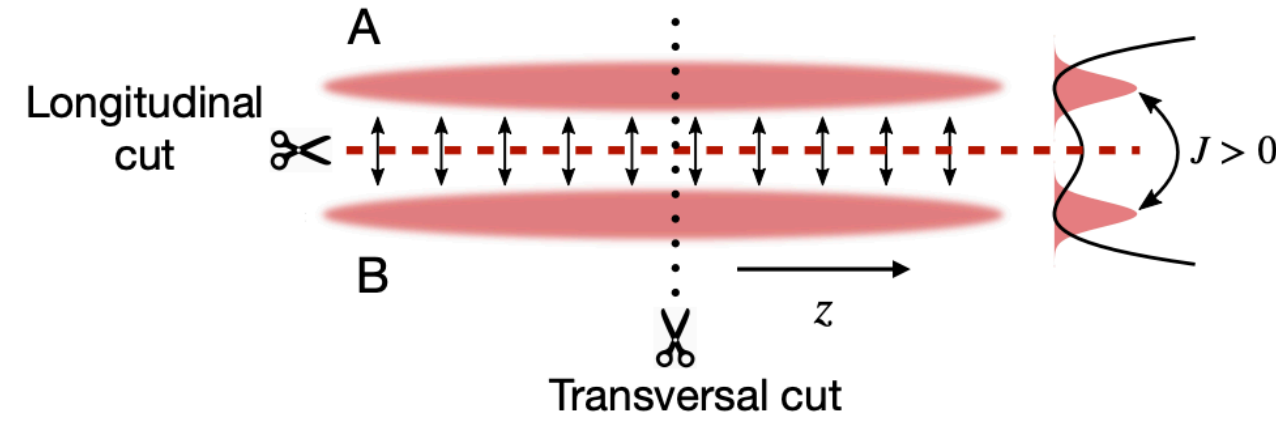
Hofferberth, S., Lesanovsky, I., Schumm, T., Imambekov, A., Gritsev, V., Demler, E., & Schmiedmayer, J. *Nat. Phys.* , 4(6), 489-495 (2008).

All correlation functions probed for relative phase are now accessible for the common phase!

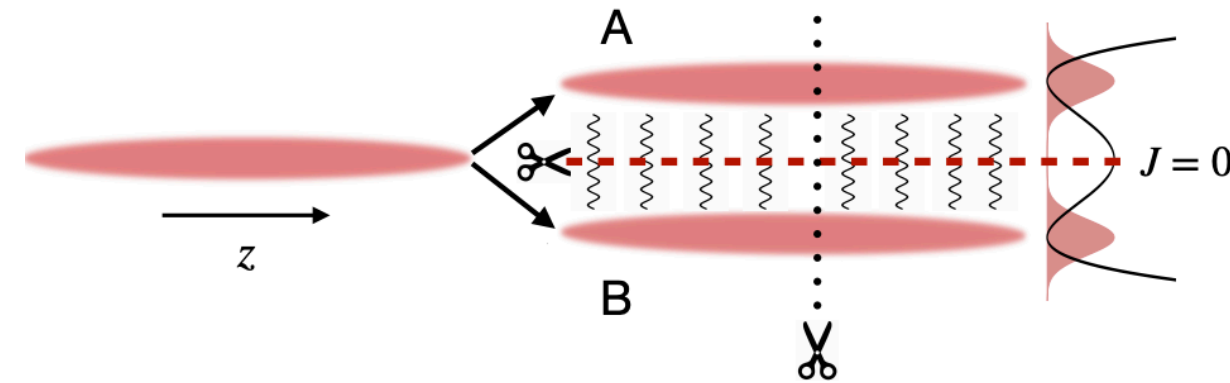
Entanglement **between** parallel 1D Bose Gases

Murtadho, T., Gluza, M., & Ng, N. H. Y. *Quantum Sci. Technol.* 11: 035060 (2026)

(a) Tunnel coupling



(b) Coherent splitting



- * Are the phonons in A entangled with phonons in B?
- * Entanglement in (Gaussian) mixed states, we use **logarithmic negativity (LN)** to characterize entanglement
- * Total correlation → **Mutual information (MI)**

Logarithmic Negativity $E_{\mathcal{N}}$

Mutual Information $I(A : B)$

Tunnel coupling

$$E_{\mathcal{N}} = \sum_{k>0}^{\Lambda} \max \left\{ 0, -\log \left[\sqrt{C_{k,J}(1 + 2\eta_k^+)(1 + 2\eta_k^-)} \right] \right\}$$

$$I(A : B) = \sum_{k>0}^{k_{\Lambda}} 2 \left[\left(\lambda_{k,J} + \frac{1}{2} \right) \log \left(\lambda_{k,J} + \frac{1}{2} \right) - \left(\lambda_{k,J} - \frac{1}{2} \right) \log \left(\lambda_{k,J} - \frac{1}{2} \right) \right] - \sum_{a=\pm} \sum_{k>0}^{k_{\Lambda}} [(\eta_k^a + 1) \log (\eta_k^a + 1) - \eta_k^a \log \eta_k^a], \quad (25)$$

Coh. Splitting

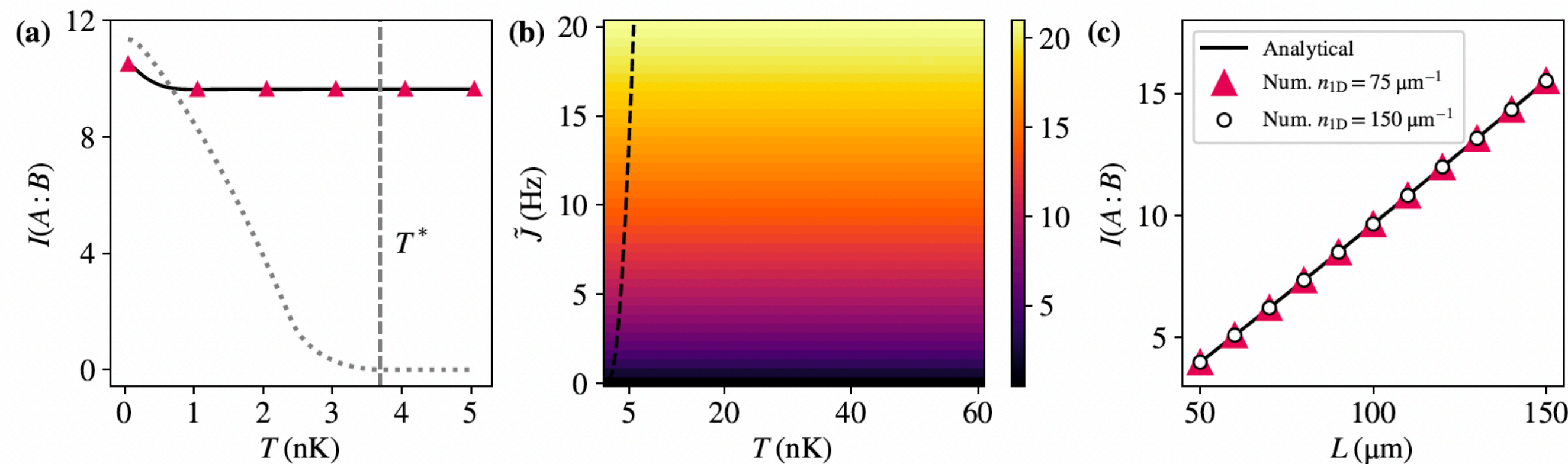
$$E_{\mathcal{N}} = \sum_{a=\pm 1} \sum_{k>0}^{\Lambda} \max \left\{ 0, -\log \left[\sqrt{C_{k,r}^a(1 + 2\eta_k^+)} \right] \right\}$$

$$I(A : B) = \sum_{k>0}^{k_{\Lambda}} 2 \left[\left(\lambda_{k,r} + \frac{1}{2} \right) \log \left(\lambda_{k,r} + \frac{1}{2} \right) - \left(\lambda_{k,r} - \frac{1}{2} \right) \log \left(\lambda_{k,r} - \frac{1}{2} \right) \right] - (1 + \eta_k^+) \log(1 + \eta_k^+) + \eta_k^+ \log \eta_k^+, \quad (31)$$

How “cold” is too “hot” for entanglement?

Murtadho, T., Gluza, M., & Ng, N. H. Y. *Quantum Sci. Technol.* 11: 035060 (2026)

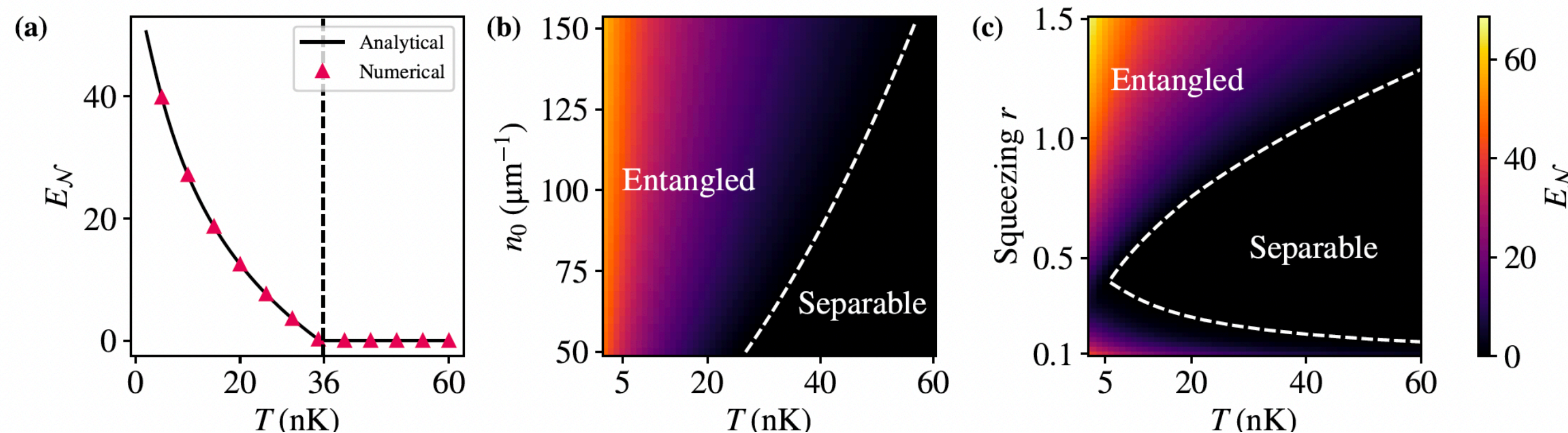
Tunnel-coupled



✱ Entanglement “sudden death”, MI constant $\rightarrow$ Quantum corr. transmutes into classical corr. (emergence of locality)

✱ Entanglement/MI scales extensively with length, but not with density $\rightarrow$ collective effect

Coherently-split



$$T^* = \sup_{0 < k \leq k_\Lambda} \left\{ T_k^* = \frac{\varepsilon_k^+}{k_B} \left(\ln \left[\sqrt{\frac{C_{k,r} + C_{k,r}^{-1} + 2}{C_{k,r} + C_{k,r}^{-1} - 2}} \right] \right)^{-1} \right\}.$$

Entanglement can be detected at temperatures accessible in the current experiments!

Entanglement Reconstruction with (Gaussian) tomography

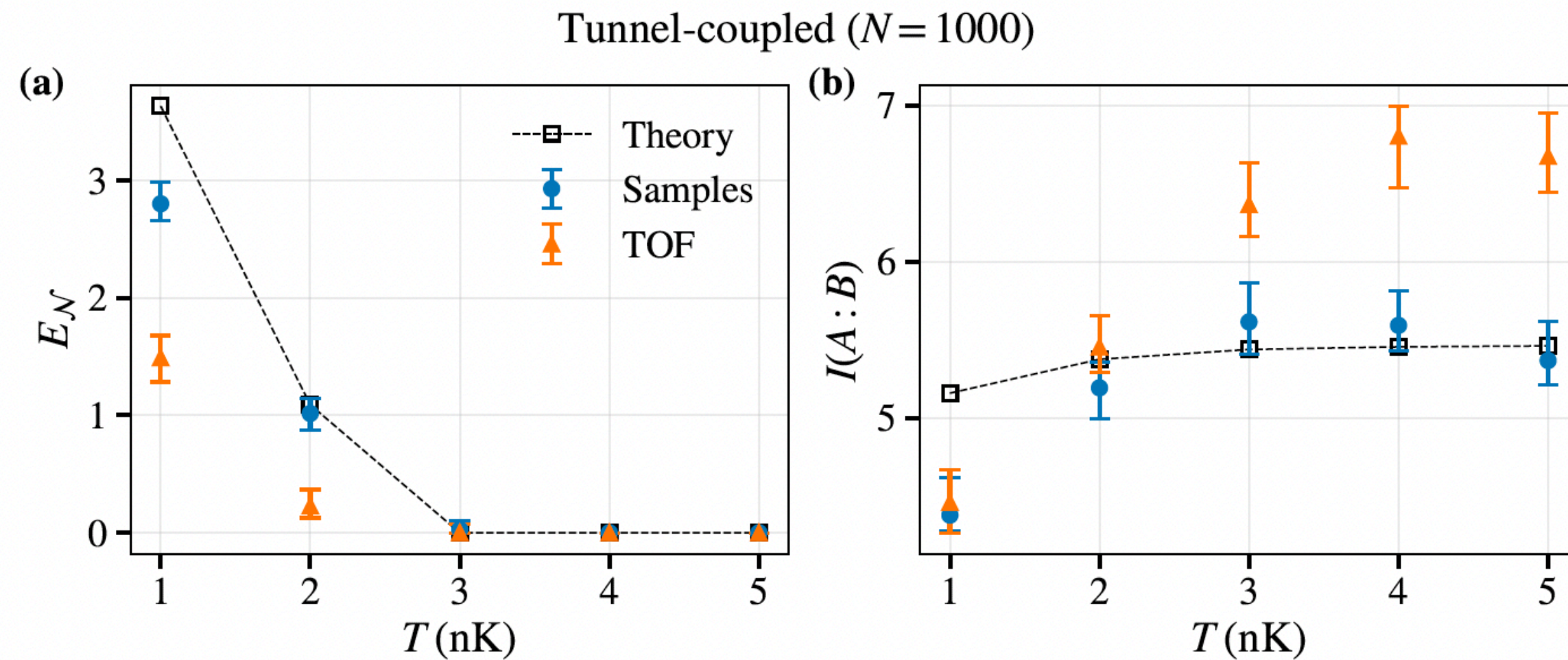


FIGURE 6.7: Tomographic reconstruction of logarithmic negativity (LN) and mutual information (MI) for tunnel-coupled gases in equilibrium. (a) LN vs.

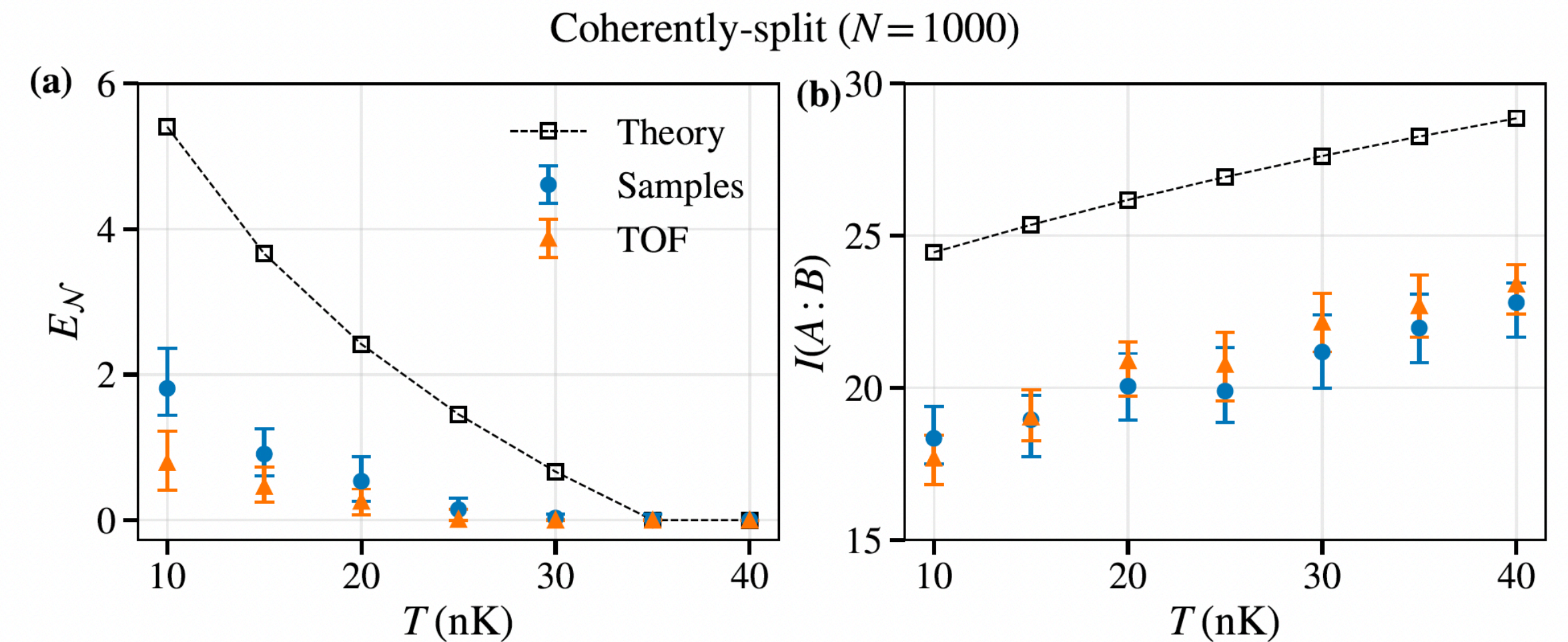


FIGURE 6.8: The same as Fig. 6.7 except that the theoretical covariance matrix is set to be that of after coherent splitting.

$$\{\phi_{\pm}(z, t)\} \rightarrow \Gamma(t) \rightarrow \begin{cases} E_{\mathcal{N}} \\ I(A : B) \end{cases}$$

**More details
coming soon!**

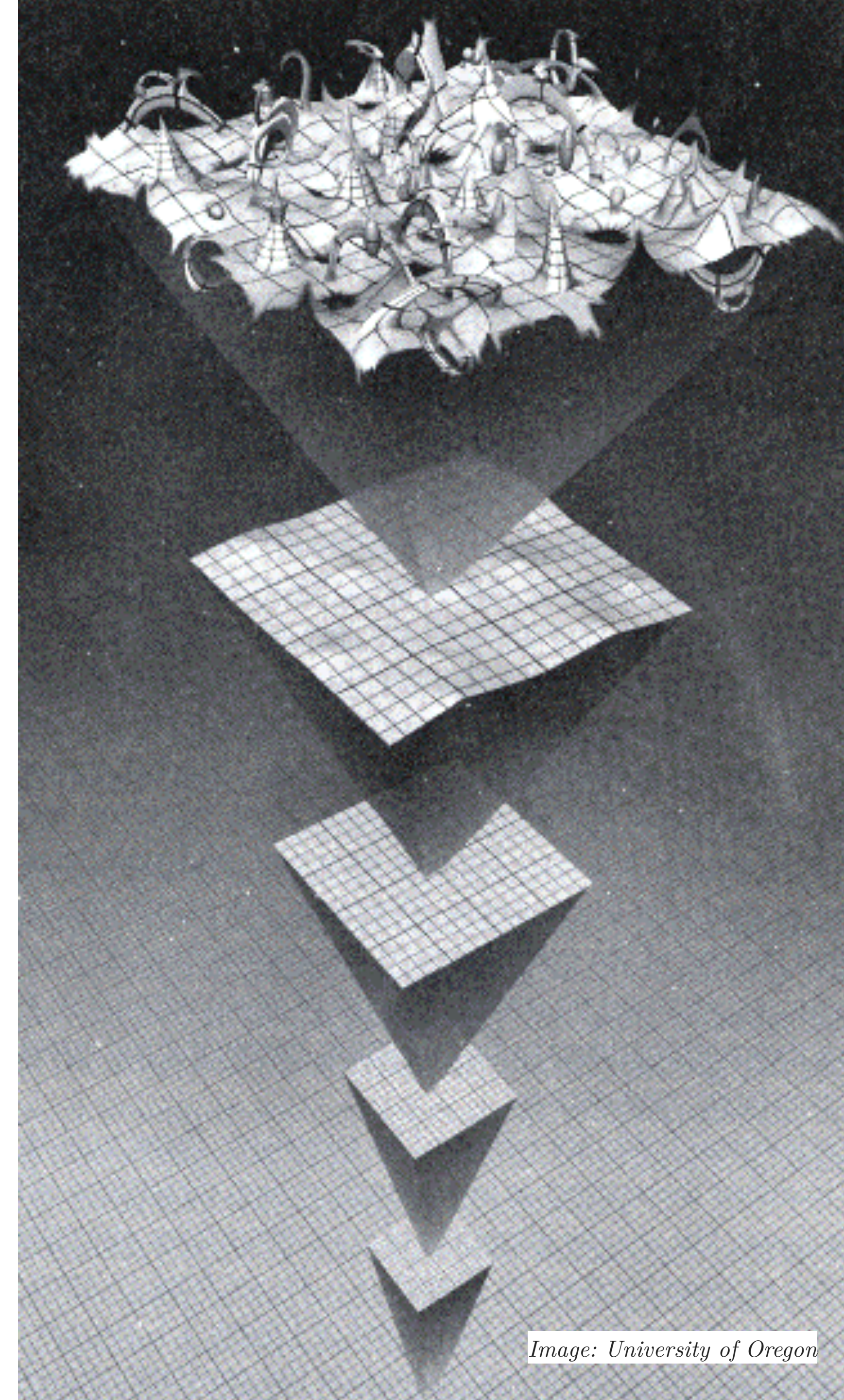
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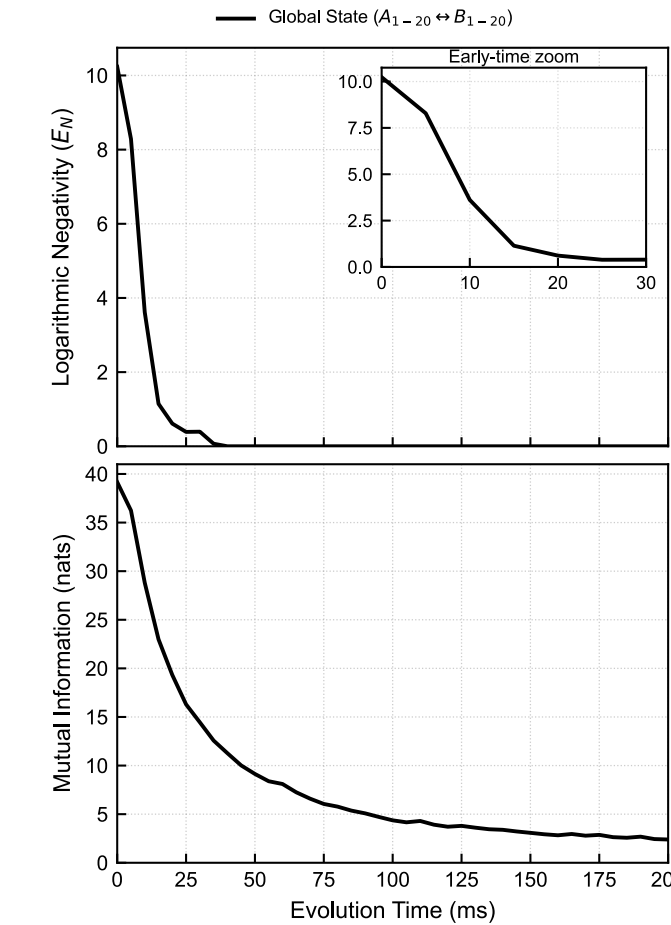
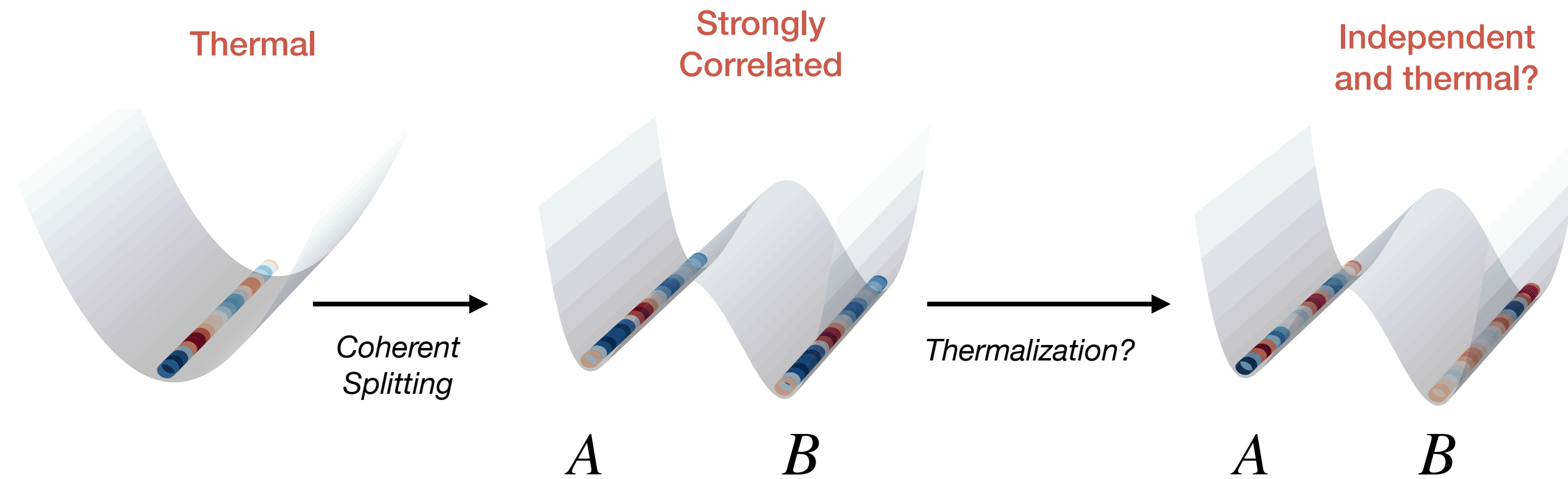
Other Applications & Take-home Messages



Other applications of common phase readout

Gring, M., et al. Science 337.6100, 1318-1322 (2012).

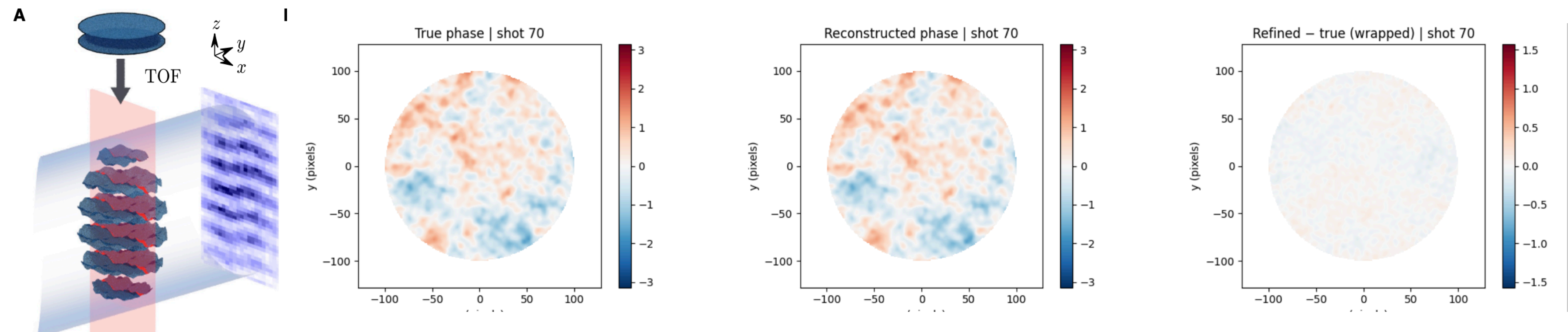
Langen, T., et al. Science 348.6231, 207-211 (2015).



Thermalization and entanglement decay in quantum near-integrable system

Rydow, E., et al. Nature Communications 16.1, 7201 (2025).

Sunami, Shinichi, et al. PRL 128.25, 250402 (2022).



Extension to 2D full phase readout

Take-home messages

Other than relative phase, we can also now measure common phase in parallel low-dimensional Bose gases

State-of-the-art experiment can already probe quantum field entanglement at finite temperature with coherent splitting

There are more applications. We live in an exciting era for quantum field simulation!

Theory: Marek Gluza, Nelly Ng

Experiment: Jörg Schmiedmayer, Sebastian Erne,
Federica Cataldini, Mohammadamin Tajik

